

Entanglement with zero distillable secret key in local dimension ten

OpenAI

September 27, 2026

Abstract

We construct an entangled state on $\mathbb{C}^{10} \otimes \mathbb{C}^{10}$ with zero distillable secret key for the local-instrument protocols specified here. They allow joint processing of all copies and unlimited two-way public communication, with the input as the only shared private resource and an eavesdropper holding a purification and the complete public record. The construction also disproves the unrestricted two-map PPT-composition conjecture. A separate trace-preserving PPT channel on $M_{21}(\mathbb{C})$ has a square that is not entanglement breaking.

1 Introduction

Secret-key distillation asks whether two distant laboratories can turn a shared noisy resource into identical random bits that an eavesdropper cannot predict. For a bipartite quantum state ρ_{AB} , Alice and Bob act locally and exchange authenticated public classical messages. Eve holds a purification of the input and receives the complete public transcript. An ideal key of length ℓ is uniform, agrees at the two laboratories, and is independent of Eve: its joint state has the form

$$2^{-\ell} \sum_{z \in \{0,1\}^\ell} |z, z\rangle \langle z, z| \otimes \sigma_{E'}.$$

We use the usual ideal-key security criterion [31, 53]. The input is the only shared private resource. Alice and Bob may process all copies jointly in their respective laboratories and use local auxiliaries and private randomness. They may also begin with a public random variable, independent of the input and its reference and recorded for Eve. Conditional on that public variable, the input-reference state and the two private auxiliary resources form a product, with fixed preparation kernels for every input.

An *admitted* protocol has a finite or $\mathbb{N}$ -indexed sequence of measurable local-instrument steps, after any supplied local aggregation. Its histories and terminal readouts must satisfy the reference-compatibility conditions of Definition 4.1. These include ordinary finite protocols with finite classical records obeying the preceding resource rules, as well as the specified countable processes. Eve retains the initial purification and the full public record, including speaker choices and stopping information. A protocol is *completed on the actual input* when both laboratories produce their prescribed outputs almost surely and the joint output is normalized.

The distillable secret key $K_D(\rho)$ is the supremum of $\liminf_n \ell_n/n$ over admitted completed key-output sequences on $\rho^{\otimes n}$ whose error from an ideal key tends to zero (Definition 5.1). All errors use the unhalved trace norm $\|X\|_1 = \text{tr} \sqrt{X^* X}$.

We construct an entangled state for which even one approximately secret bit is uniformly out of reach. A density operator is a positive matrix of trace one. It is *separable* if it is a finite sum of positive product matrices, and *entangled* otherwise.

Theorem 1.1. *There is an explicitly specified entangled density operator ρ on $\mathbb{C}^{10} \otimes \mathbb{C}^{10}$ whose range contains no nonzero product vector and for which $K_D(\rho) = 0$. More precisely, let $n \geq 1$ and let $\tau_{K_A K_B E'}$ be the normalized output of any admitted completed bit-output protocol of Definition 4.1 on $\rho^{\otimes n}$, where E' retains an initial purification and the complete public transcript. For every state $\sigma_{E'}$,*

$$\left\| \tau_{K_A K_B E'} - \frac{1}{2} \sum_{i=0}^1 |i, i\rangle \langle i, i| \otimes \sigma_{E'} \right\|_1 \geq \frac{1}{5}. \quad (1.1)$$

Thus the corresponding trace distance, with its conventional factor $1/2$, is at least $1/10$.

There is no uniform bound on the amount of communication or the number of steps within the admitted model. Proposition 5.4 also gives separate consequences for public conditioning on an event of positive probability, vanishing abort probability, and trace-norm limits of completed outputs whose Eve registers retain or allow recovery of the purification and complete public record.

The state comes from a construction for completely positive maps. A complex-linear map $F : M_a(\mathbb{C}) \rightarrow M_b(\mathbb{C})$ is *completely positive* (CP) when $\text{id}_k \otimes F$ preserves positive matrices for every $k \geq 1$. We call it *PPT* when both F and $T_b \circ F$ are CP, where T_b is coordinate transposition. Write $J(F)$ for the unnormalized input-first Choi matrix, defined in (2.1). A CP map is *entanglement breaking* when applying it to one subsystem of any positive bipartite input always gives a separable output. Equivalently, its Choi matrix is separable [33, Theorem 4]; this cone formulation does not require trace preservation.

Theorem 1.2. *There are explicitly specified PPT maps $\Phi_1, \Phi_2 : M_{10}(\mathbb{C}) \rightarrow M_{10}(\mathbb{C})$ for which*

$$Z = J(\Phi_2 \circ \Phi_1)$$

is nonzero and its range contains no nonzero product vector in $\mathbb{C}^{10} \otimes \mathbb{C}^{10}$. In particular, the composition $\Phi_2 \circ \Phi_1$ is not entanglement breaking.

The maps are given by (6.5), starting from the integer matrices in (7.1). They may differ, and the construction does not impose trace preservation or unitality. The state in Theorem 1.1 is

$$\rho = \frac{Z}{\text{tr } Z}. \quad (1.2)$$

The denominator is positive because Z is a nonzero positive Choi matrix. A separate construction supplies a trace-preserving example.

Theorem 1.3. *There is an explicitly specified trace-preserving PPT channel $\Theta : M_{21}(\mathbb{C}) \rightarrow M_{21}(\mathbb{C})$ whose square $\Theta \circ \Theta$ is not entanglement breaking.*

The pair in Theorem 1.2 gives a negative answer to the unrestricted two-map PPT-composition conjecture of Christandl, Müller-Hermes, and Wolf [16, Conjecture IV.1]. The channel in Theorem 1.3 answers the channel question recorded in the 2012 BIRS workshop report [54, Problem G], already with the same channel in both positions. We do not claim that either dimension is minimal.

Entanglement and secrecy. Public discussion and privacy amplification are central to the classical theory of key agreement [9, 41, 7]. Entanglement purification provides a quantum route to secrecy: nearly pure shared entangled pairs decouple from a purifying adversary [8, 11, 19]. Bound entanglement showed that entanglement need not be distillable into singlets. In particular, PPT entangled states have zero distillable entanglement [32]. Some PPT entangled states nevertheless

have positive distillable secret key [30]. Their private-state description explains how secrecy can persist in a quantum shield even when singlet distillation is impossible [31].

Separable states have zero key against a purifying Eve [17, 30]. The converse was posed explicitly as Problem 24, “Secret key from all entangled states,” in Krüger and Werner’s open-problem collection: the entry is dated 15 March 2005 and lists P. Horodecki as contact [40]. The question is also discussed in the private-state literature [31, 29]. One-way coding theorems give useful positive-key criteria [20], while Theorem 1.1 in particular rules out a positive key rate for a specific entangled state under ordinary finite two-way protocols, whose round count may grow with the input length. The distinction between secrecy needed to prepare correlations and key recoverable from them also matters: measurements of an entangled state can yield correlations with a positive secrecy formation cost without proving a positive distillation rate [26, 25, 1].

There are earlier quantitative obstructions near private bits. Kim and Sanders obtained a distance bound for PPT states under a Hermitian-block hypothesis [38], and Badziąg and coauthors removed that hypothesis [5]. Those bounds depend on the shield dimension, which may grow with a protocol. Our bound applies to the final classical bit registers and Eve’s complete public side information, with a constant independent of the input length and the local memories.

Pauwels, Gisin, and Renner recently constructed a finite classical bipartite source with positive secrecy formation cost and zero asymptotic secret-key rate under finite-round authenticated public discussion, with the round count allowed to grow with block length [46, Theorem 1 and Appendix A]. Their revised paper also proves positive two-way distillable quantum key for the Horodecki two-qutrit states σ_α with $3 < \alpha \leq 5$, including the bound-entangled range $3 < \alpha \leq 4$, against a purifying quantum Eve [46, Corollary 3 and Appendix I.7]. The state in Theorem 1.1 gives the quantum zero-key conclusion under the local quantum protocol model above.

The operational obstruction. The first part of the proof concerns a class $\mathcal{C}$ of states ω on $\mathbb{C}^a \otimes \mathbb{C}^b$ with a local representation

$$\omega = (\mathcal{L} \otimes \mathcal{M})(\chi\chi^*), \quad \chi \in \mathbb{C}^r \otimes \mathbb{C}^s, \quad \mathcal{L} : M_r(\mathbb{C}) \rightarrow M_a(\mathbb{C}), \quad \mathcal{M} : M_s(\mathbb{C}) \rightarrow M_b(\mathbb{C}),$$

where $\mathcal{L}, \mathcal{L} \circ T_r, \mathcal{M}, \mathcal{M} \circ T_s$ are CP. These input-transpose conditions are equivalent to the output-transpose PPT convention above, as Lemma 2.1 shows. The maps may be rectangular and need not preserve trace; only the output ω has trace one. This condition persists under tensor powers. Relative to one common transcript measure, the four final bit branches have positive local effect densities $X_i(t) \otimes Y_j(t)$ on the two local input spaces of the protocol for almost every complete transcript t , with the same two local pairs for all $i, j \in \{0, 1\}$. To prove this, we run the protocol on auxiliary product probes. Their two local sampling tapes are conditionally independent given the complete public record. Informationally complete tests on untouched references turn that independence into common product factors, and a filter on the references transfers the factors to the actual input.

For a state in $\mathcal{C}$, a coherence inequality bounds the overlap between the 00 and 11 branches by the probabilities of disagreement. After integration over transcripts, let e be the total probability of different bits and let $\mathfrak{f}$ be the root fidelity between Eve’s unnormalized 00 and 11 conditional states, summed over the public record. We prove

$$\mathfrak{f} \leq \sqrt{e(1-e)} + e.$$

If the output is at trace-norm distance η from an ideal bit, correctness and secrecy give, respectively,

$$e \leq \frac{\eta}{2}, \quad \mathfrak{f} \geq \frac{1-\eta}{2}.$$

For $\eta < 1/5$, the right side of the first display is less than $2/5$, whereas the second display requires more than $2/5$. This proves the uniform gap. Taking the first bit of a longer ideal key is a trace-norm contraction, so no positive asymptotic key rate is possible.

How the dimension-ten construction enters. For CP maps $F : M_a(\mathbb{C}) \rightarrow M_b(\mathbb{C})$ and $G : M_b(\mathbb{C}) \rightarrow M_c(\mathbb{C})$, define $F^\sharp = T_a \circ F^\dagger \circ T_b$, where $F^\dagger$ is the Hilbert–Schmidt adjoint. The transpose-adjoint Choi identity is

$$J(G \circ F) = (F^\sharp \otimes G)(\Omega_b \Omega_b^*), \quad \Omega_b = \sum_{i=1}^b e_i \otimes e_i.$$

This identity and the local-product formulation are established machinery [16, Lemma I.1(2), Section IV.B]; we prove the needed rectangular form and normalization in Lemma 2.6. When F and G are PPT and the Choi matrix is nonzero, division by its trace gives a state in $\mathcal{C}$. Applying this observation to Theorem 1.2 proves that (1.2) belongs to the class covered by the operational obstruction.

It remains to make this state entangled. A nonzero separable positive matrix has a nonzero product vector in its range, a necessary part of the range criterion [34]. We exclude such vectors directly. Start with a PPT map $L : M_4(\mathbb{C}) \rightarrow M_4(\mathbb{C})$ and twenty projective directions $[x]$ at which $L(xx^*)$ is singular. No nonzero homogeneous quadratic in four variables vanishes at ten of these directions. The ten-dimensional symmetric square records the quadratic monomials of x , and the six-dimensional exterior square converts rank loss into a product vector in $\ker Z$. If $\text{ran } Z$ contained a product vector, its two factors would give two nonzero quadratics whose zero sets cover all twenty directions. Each can cover at most nine, and $9 + 9 < 20$. A separate positive definite output of L ensures that Z is nonzero.

This range argument is related to unextendible product bases [10], nonorthogonal product exclusion by linear independence [49, Section IV, Lemma 3], and polynomial evaluation methods [45, Proposition 1.3]. The complementary-pair operation on the exterior square agrees with the four-dimensional fermionic duality of Schliemann, Loss, and MacDonald [55, Section III]. Our explicit L comes from a real integer 6-by-4 matrix pencil. Its rank-losing directions solve a bilinear system of the type appearing in the study of low-rank PPT states on $\mathbb{C}^4 \otimes \mathbb{C}^4$ [27, Section 7]. Exact reductions at three primes prove that the relevant degree-twenty algebra is a field with normal-closure Galois group S_{20} . This symmetry carries one nonzero quadratic evaluation determinant to every ten-point choice. The arithmetic certificate and its precise computational coverage are given in Appendix A.

The channel and the projection construction. The PPT-composition problem has important affirmative cases, including two maps on $M_3(\mathbb{C})$ [16, Corollary III.1]; see also [12]. Other cases include Gaussian channels [16] and diagonal unitary covariant or conjugate diagonal unitary covariant PPT maps [56, Theorem 4.5]. Asymptotic approach to the entanglement-breaking cone was proved for iterates of unital or trace-preserving PPT maps [37, Theorem 3.3]. Finite entanglement-breaking index was proved for unital PPT channels by Rahaman, Jaques, and Paulsen [52, Theorem 4.4], and for PPT channels with a full-rank invariant state by Hanson, Rouzé, and Stilck França [28, Theorem 3.10]. More recent results concern different scopes. An and Lee prove that a qutrit CP map whose Choi matrix is one-copy undistillable and a qutrit CP map whose Choi matrix has Schmidt number at most two compose to an entanglement-breaking map in either order [3, Theorem 4]. Park’s preprint states eventual entanglement breaking for each CP and completely copositive endomap, with a map-dependent number of iterations [44, Theorem 1.1]. A failure at the second iterate is compatible with such eventual behavior.

To obtain Theorem 1.3, we rescale the two maps, alternate them between two coordinate blocks, and send the missing trace to an absorbing flag. The switch follows Christandl, Müller-Hermes, and Wolf [16, Section IV.A]; the flag completion follows Filippov’s trace-preserving extension [21, Proposition 3, Equation (15)]. The Choi matrix of the square has a local corner equal to a positive multiple of Z . Local compression preserves separability, so that corner certifies that the square is not entanglement breaking.

The final part of the paper develops a second way to retain entanglement under two PPT maps. A rectangular criterion starts from real unit vectors and projective measurements, uses a block Gram matrix for positivity, and uses commutation for the partial-transpose property. A support argument using zero tensor-basis diagonal entries rules out separability over the complex field. We realize the criterion by a restricted-intersection construction [23, 2] and tensor products of the Mermin–Peres parity measurements [42, 47, 48]. Its dimensions are much larger, but the mechanism is independent of the twenty-direction geometry. Frankl and Rödl’s forbidden-intersection theory similarly forces fixed-size orthogonal families in large subsets of a sign cube [22, Theorem 1.11]; our eight-point estimate is proved by a direct polynomial bound and an elementary tensor-power fiber argument. The real construction also gives an entangled output when the same trace-preserving rectangular PPT channel is applied to both halves of a maximally entangled state. This distinction between real matrices and complex separability matters: the real-field example of Chiribella, Davidson, Paulsen, and Rahaman has an entanglement-breaking complexification [13, Example 9.6].

Organization. Section 2 establishes the common map conventions, represented class, and Choi transfer. Sections 3–5 prove the operational theorem. Sections 6 and 7 construct the dimension-ten pair, and Section 8 proves the state and channel consequences. Sections 9 and 10 give the independent projection and combinatorial method. The appendices give the exact certificate, the general algebraic facts used in its proof, and the supplementary finite checks of the explicit data.

2 PPT maps and the represented class

All matrix algebras in the constructions are finite-dimensional over $\mathbb{C}$, with fixed orthonormal bases and the induced product bases. We write X^* for conjugate transpose, $\overline{X}$ for entrywise conjugation, and $T_d(X) = X^\top$ for the complex-linear transpose on $M_d(\mathbb{C})$. For a linear map F , its Hilbert–Schmidt adjoint $F^\dagger$ is characterized by $\text{tr}(Y^*F(X)) = \text{tr}((F^\dagger(Y))^*X)$. If V is a matrix, then $\text{Ad}_V(X) = V X V^*$.

For $F : M_a(\mathbb{C}) \rightarrow M_b(\mathbb{C})$ our Choi convention is

$$J(F) = \sum_{i,j=1}^a E_{ij} \otimes F(E_{ij}) = (\text{id}_a \otimes F)(\Omega_a \Omega_a^*), \quad \Omega_a = \sum_{i=1}^a e_i \otimes e_i. \quad (2.1)$$

In particular, $\|\Omega_a\|^2 = a$; the vector is not normalized. The first tensor factor is the input reference. This convention fixes every dimension factor in the transfer and channel formulas below. It is the map–operator correspondence associated with Jamiołkowski and Choi [35, 15].

Lemma 2.1. *Let $F : M_a(\mathbb{C}) \rightarrow M_b(\mathbb{C})$ be CP. Then $F^\dagger$ and $T_b \circ F \circ T_a$ are CP. Moreover,*

$$F \circ T_a \text{ is CP} \iff T_b \circ F \text{ is CP}.$$

If F is PPT, so are $T_b \circ F$, $F \circ T_a$, its adjoint, its simultaneous transpose $T_b \circ F \circ T_a$, and its transpose-adjoint $F^\sharp := T_a \circ F^\dagger \circ T_b$. Tensor products of PPT maps are PPT. If P and Q are compatible CP maps, then $Q \circ F \circ P$ is PPT whenever F is PPT.

Proof. Choose Kraus matrices $A_r \in \mathbb{C}^{b \times a}$ such that $F(X) = \sum_r A_r X A_r^*$. Then

$$F^\dagger(Y) = \sum_r A_r^* Y A_r, \quad (T_b \circ F \circ T_a)(X) = \sum_r \overline{A_r} X \overline{A_r}^*,$$

which proves both CP assertions. Applying the simultaneous-transpose assertion to $F \circ T_a$ or to $T_b \circ F$ proves their equivalence. This argument works with rectangular Kraus matrices. The operator-sum representation is standard [39, 15].

Transpose is self-adjoint for the Hilbert–Schmidt inner product. Hence, when F is PPT,

$$T_a \circ F^\dagger = (F \circ T_a)^\dagger \quad \text{and} \quad F^\dagger \circ T_b = (T_b \circ F)^\dagger$$

are CP. This proves the PPT assertion for the adjoint, and simultaneous transposition gives the simultaneous transpose and transpose-adjoint. The output transpose of $T_b \circ F$ is F , while that of $F \circ T_a$ is $T_b \circ F \circ T_a$; all these maps are CP. This proves the remaining PPT assertions. With product bases, $T_{b_1 b_2} = T_{b_1} \otimes T_{b_2}$, so the tensor-product assertion follows from closure of CP maps under tensor products. Finally, for $P : M_{a'}(\mathbb{C}) \rightarrow M_a(\mathbb{C})$ and $Q : M_b(\mathbb{C}) \rightarrow M_{b'}(\mathbb{C})$,

$$T_{b'} \circ Q \circ F \circ P = (T_{b'} \circ Q \circ T_b) \circ (T_b \circ F) \circ P$$

is a composition of CP maps. The untransposed composition is CP as well. $\square$

The input- and output-transpose formulations of PPT will therefore be used interchangeably. No trace condition is included in this terminology. The elementary closure facts above are also recorded, in related forms, in [37, Lemma 2.1].

Lemma 2.2. *A linear map $F : M_a(\mathbb{C}) \rightarrow M_b(\mathbb{C})$ is CP if and only if $J(F) \succeq 0$. For a CP map, the following are equivalent:*

1. $J(F)$ is separable;
2. for every $k \geq 1$ and every $W \succeq 0$ in $M_k(\mathbb{C}) \otimes M_a(\mathbb{C})$, $(\text{id}_k \otimes F)(W)$ is separable.

Also,

$$J(T_b \circ F) = (\text{id}_a \otimes T_b)J(F), \quad J(F \circ T_a) = (T_a \otimes \text{id}_b)J(F).$$

Proof. The forward CP implication follows from (2.1). Conversely, write a positive Choi matrix as $J(F) = \sum_r v_r v_r^*$. There is a unique matrix $A_r \in \mathbb{C}^{b \times a}$ with $v_r = (I_a \otimes A_r)\Omega_a$. Comparison of the (i, j) blocks gives $F(E_{ij}) = \sum_r A_r E_{ij} A_r^*$ and hence the Kraus representation of F . This is Choi's finite-dimensional criterion [15].

If $J(F)$ is separable, spectral decompositions of its positive factors write it as $\sum_r a_r a_r^* \otimes b_r b_r^*$. Comparison of blocks now gives the measure-and-prepare formula

$$F(X) = \sum_r \text{tr}[(a_r a_r^*)^\top X] b_r b_r^*.$$

Put $E_r = (a_r a_r^*)^\top \succeq 0$. For $W \succeq 0$, each partial matrix

$$\text{tr}_a[(I_k \otimes E_r)W] = \text{tr}_a[(I_k \otimes \sqrt{E_r})W(I_k \otimes \sqrt{E_r})]$$

is positive. Thus $(\text{id}_k \otimes F)(W)$ is a sum of positive products with second factors $b_r b_r^*$. The converse follows by applying the asserted property to $W = \Omega_a \Omega_a^*$. This proves the entanglement-breaking characterization [33, Theorem 4] at the level of the CP cone. The two transpose identities follow directly by reindexing the matrix units in (2.1). $\square$

Lemma 2.3. *Every nonzero separable positive matrix has a nonzero product vector in its range. Consequently, a nonzero positive matrix whose range contains no nonzero product vector is nonseparable.*

Proof. Refine a separable decomposition by the spectral decompositions of its positive factors. It then has the form $Z = \sum_r v_r v_r^*$ with product vectors $v_r = a_r \otimes b_r$. Since $\ker Z = \bigcap_r v_r^\perp$, every v_r lies in $(\ker Z)^\perp = \text{ran } Z$. If $Z \neq 0$, at least one v_r is nonzero. $\square$

This is the range condition we use from the range criterion [34]. It is only a necessary condition for separability; the dimension-ten construction below establishes its failure directly.

Definition 2.4. A density operator ω on $\mathbb{C}^a \otimes \mathbb{C}^b$ belongs to the represented class $\mathcal{C}$ if there are positive integers r, s , a vector $\chi \in \mathbb{C}^r \otimes \mathbb{C}^s$, and linear maps $\mathcal{L} : M_r(\mathbb{C}) \rightarrow M_a(\mathbb{C})$ and $\mathcal{M} : M_s(\mathbb{C}) \rightarrow M_b(\mathbb{C})$ such that

$$\omega = (\mathcal{L} \otimes \mathcal{M})(\chi\chi^*), \quad \mathcal{L}, \mathcal{L} \circ T_r, \mathcal{M}, \mathcal{M} \circ T_s \text{ are CP.} \quad (2.2)$$

Only ω is required to have trace one. The vector χ may be unnormalized, and the maps need not preserve trace.

This is a representation property, with no assumed physical implementation of the representing maps. Lemma 2.1 shows that the two maps are PPT also in the output convention. In particular the state itself has positive partial transpose, since, for example, $(T_a \otimes \text{id}_b)(\omega) = ((T_a \circ \mathcal{L}) \otimes \mathcal{M})(\chi\chi^*) \succeq 0$. The operational proof will use the representing maps themselves, in addition to this partial-transpose consequence.

Lemma 2.5. *The class $\mathcal{C}$ is closed under finite tensor products, after grouping all first-party systems and all second-party systems. In particular, $\omega^{\otimes n} \in \mathcal{C}$ for every $\omega \in \mathcal{C}$ and every $n \geq 1$.*

Proof. Tensor the representing vectors and the first-party maps separately from the second-party maps, and use the canonical local permutation unitaries to group the systems. Tensor products of CP maps are CP, and input transpose in the grouped product basis is the tensor product of the individual input transposes. Hence the four maps required in (2.2) remain CP. The output is the tensor product of the original states and has trace one. The local permutation unitaries and their conjugates are CP pre- and post-compositions, so they do not affect the conclusion. $\square$

Lemma 2.6 (Choi transfer). *Let $F : M_a(\mathbb{C}) \rightarrow M_b(\mathbb{C})$ and $G : M_b(\mathbb{C}) \rightarrow M_c(\mathbb{C})$ be CP maps, and set $F^\sharp = T_a \circ F^\dagger \circ T_b$. Then*

$$J(G \circ F) = (F^\sharp \otimes G)(\Omega_b \Omega_b^*). \quad (2.3)$$

If F and G are PPT and $Z = J(G \circ F) \neq 0$, then $\lambda := \text{tr } Z > 0$ and

$$\frac{Z}{\lambda} = (F^\sharp \otimes G) \left(\frac{\Omega_b \Omega_b^*}{\lambda} \right) \in \mathcal{C} \quad \text{on } \mathbb{C}^a \otimes \mathbb{C}^c. \quad (2.4)$$

Proof. Let $A_r \in \mathbb{C}^{b \times a}$ and $B_s \in \mathbb{C}^{c \times b}$ be Kraus matrices for F and G . Transposing the adjoint formula gives

$$F^\sharp(Y) = \sum_r A_r^\top Y \overline{A_r} = \sum_r A_r^\top Y (A_r^\top)^*.$$

Thus the Kraus matrices of $F^\sharp : M_b(\mathbb{C}) \rightarrow M_a(\mathbb{C})$ are the transposes $A_r^\top$. For each pair of Kraus matrices, direct expansion in the fixed bases gives

$$(A_r^\top \otimes B_s) \Omega_b = (I_a \otimes B_s A_r) \Omega_a.$$

Summing the corresponding rank-one matrices proves (2.3). This is the rectangular CP form of the transpose-adjoint identity in [16, Lemma I.1(2)]; its use for local products of PPT maps is discussed there in Section IV.B.

Suppose now that F and G are PPT. Besides complete positivity of $F^\sharp$ and G , Lemma 2.1 gives

$$F^\sharp \circ T_b = T_a \circ F^\dagger = (F \circ T_a)^\dagger \text{ CP}, \quad G \circ T_b \text{ CP}.$$

These are exactly the input-transpose conditions for the two local maps in Definition 2.4. Finally,

$$Z = \sum_{r,s} |(I_a \otimes B_s A_r) \Omega_a\rangle \langle (I_a \otimes B_s A_r) \Omega_a|, \quad \lambda = \sum_{r,s} \|B_s A_r\|_F^2.$$

The nonzero positive matrix Z has $\lambda > 0$. Taking $\chi = \Omega_b / \sqrt{\lambda}$ in (2.2) proves (2.4). Its squared norm is b/λ , which is allowed by the definition. No trace-preserving hypothesis or additional factor of b is used. $\square$

3 A four-effect coherence bound

The first step in the operational argument is an inequality for four arbitrary positive local matrices. It compares the coherence between two equal-bit branches with the weights of branches in which the two bits differ. The matrices need not form measurements: later they will be densities at a fixed public transcript, where pointwise normalization is not required.

We use the represented class $\mathcal{C}$ of Definition 2.4. Thus, if $\omega \in \mathcal{C}$ acts on $\mathbb{C}^p \otimes \mathbb{C}^q$, then for some positive integers r, s it has a representation

$$\omega = (\mathcal{L} \otimes \mathcal{M})(\chi \chi^*), \quad \chi \in \mathbb{C}^r \otimes \mathbb{C}^s,$$

where $\mathcal{L} : M_r(\mathbb{C}) \rightarrow M_p(\mathbb{C})$ and $\mathcal{M} : M_s(\mathbb{C}) \rightarrow M_q(\mathbb{C})$ satisfy that $\mathcal{L}, \mathcal{L} \circ T_r, \mathcal{M}, \mathcal{M} \circ T_s$ are completely positive. Only ω has trace one; the vector and the rectangular maps have no normalization requirement. We will use that freedom to change coordinates on the input of $\mathcal{L}$.

Lemma 3.1 (Positive block estimate). *Let H and J be square matrices, possibly of different sizes, and let Z have the corresponding rectangular size. If*

$$\begin{pmatrix} H & Z \\ Z^* & J \end{pmatrix} \succeq 0,$$

then

$$\|Z\|_1 \leq \sqrt{\text{tr } H \text{ tr } J}.$$

This estimate includes singular H and J .

Proof. A spectral decomposition of the positive block matrix expresses it as a finite sum of outer products of block columns $\begin{pmatrix} a_k \\ b_k \end{pmatrix}$. Consequently $H = \sum_k a_k a_k^*$, $J = \sum_k b_k b_k^*$, and $Z = \sum_k a_k b_k^*$. Since a rank-one matrix ab^* has trace norm $\|a\| \|b\|$, the triangle inequality and Cauchy–Schwarz give

$$\|Z\|_1 \leq \sum_k \|a_k\| \|b_k\| \leq \left(\sum_k \|a_k\|^2 \right)^{1/2} \left(\sum_k \|b_k\|^2 \right)^{1/2} = \sqrt{\text{tr } H \text{ tr } J}.$$

No inverse of a diagonal block is used. $\square$

Lemma 3.2 (Four-effect inequality). *Let $\omega \in \mathcal{C}$ act on $\mathbb{C}^p \otimes \mathbb{C}^q$. For arbitrary $X_0, X_1 \succeq 0$ in $M_p(\mathbb{C})$ and $Y_0, Y_1 \succeq 0$ in $M_q(\mathbb{C})$, put*

$$D_{ij} = X_i \otimes Y_j, \quad p_{ij} = \text{tr}(D_{ij}\omega), \quad i, j \in \{0, 1\}.$$

Then

$$\left\| \sqrt{D_{00}} \omega \sqrt{D_{11}} \right\|_1 \leq \sqrt{p_{00}p_{01}} + \sqrt{p_{10}p_{11}} + 2\sqrt{p_{01}p_{10}}. \quad (3.1)$$

There is no normalization assumption on X_0, X_1, Y_0, Y_1 .

Proof. Fix a representation of ω as above. Pull the four matrices back to the two input spaces by the Hilbert–Schmidt adjoints:

$$A_i = \mathcal{L}^\dagger(X_i) \in M_r(\mathbb{C}), \quad B_j = \mathcal{M}^\dagger(Y_j) \in M_s(\mathbb{C}).$$

These matrices are positive. For instance, for every $a \in \mathbb{C}^r$,

$$a^* A_i a = \text{tr}[X_i \mathcal{L}(aa^*)] \geq 0.$$

We first replace the representation by one in which A_0 and A_1 commute; the two complete-positivity conditions will then give complementary estimates for each off-diagonal term of the input.

Normalization on the support. Set $H = A_0 + A_1$, and let Π be the orthogonal projection onto $\text{ran } H$. If $a \in \ker H$, the nonnegative numbers $a^* A_i a = \|\sqrt{A_i} a\|^2$ sum to zero, so each A_i annihilates $\ker H$. It follows also that $A_i = \Pi A_i \Pi$. Define S to be $H^{-1/2}$ on $\text{ran } H$ and the identity on $\ker H$. The matrix S is positive and invertible on all of $\mathbb{C}^r$, including when $H = 0$. The two positive matrices

$$A'_i = S^* A_i S$$

are supported on Π and satisfy $A'_0 + A'_1 = \Pi$. Hence they commute: $[A'_0, A'_1] = [A'_0, \Pi - A'_0] = 0$.

This congruence can be made inside the representation without changing ω . With $\text{Ad}_S(U) = SUS^*$, replace

$$\mathcal{L} \quad \text{by} \quad \mathcal{L}' = \mathcal{L} \circ \text{Ad}_S, \quad \chi \quad \text{by} \quad \chi' = (S^{-1} \otimes I_s)\chi.$$

The resulting output is the same, and the pullbacks of X_i are A'_i . The new map is CP, and its input-transpose condition follows from

$$\text{Ad}_S \circ T_r = T_r \circ \text{Ad}_{\bar{S}}, \quad \mathcal{L}' \circ T_r = (\mathcal{L} \circ T_r) \circ \text{Ad}_{\bar{S}}.$$

Thus it remains a representation of the kind required by Definition 2.4. We now suppress the primes.

The two block estimates. Let $C = \mathcal{L} \otimes \mathcal{M}$. For input vectors $u, z \in \mathbb{C}^r \otimes \mathbb{C}^s$, define

$$\begin{aligned} p_{ij}(u) &= u^*(A_i \otimes B_j)u = \text{tr}[D_{ij}C(uu^*)], \\ Z(u, z) &= \sqrt{D_{00}} C(uz^*) \sqrt{D_{11}}. \end{aligned}$$

Complete positivity of C means that applying it entrywise to

$$\begin{pmatrix} uu^* & uz^* \\ zu^* & zz^* \end{pmatrix} = \begin{pmatrix} u \\ z \end{pmatrix} \begin{pmatrix} u \\ z \end{pmatrix}^*$$

gives a positive block matrix. Conjugate that matrix by $\text{diag}(\sqrt{D_{00}}, \sqrt{D_{11}})$ and apply Lemma 3.1. The result is

$$\|Z(u, z)\|_1 \leq \sqrt{p_{00}(u)p_{11}(z)}. \quad (3.2)$$

The second estimate concerns the same $Z(u, z)$, but exchanges the two input vectors in the diagonal weights. In the product input basis the *global input transpose* is $T_{rs} = T_r \otimes T_s$. The map

$$C \circ T_{rs} = (\mathcal{L} \circ T_r) \otimes (\mathcal{M} \circ T_s)$$

is CP. Apply the preceding block argument to this map, with the ordered pair $(\bar{z}, \bar{u})$ and the same two output filters. The identities

$$(\bar{z} \bar{u}^*)^\top = uz^*, \quad (\bar{z} \bar{z}^*)^\top = zz^*, \quad (\bar{u} \bar{u}^*)^\top = uu^*$$

show that the off-diagonal output is unchanged while the two diagonal inputs are exchanged. We obtain

$$\|Z(u, z)\|_1 \leq \sqrt{p_{00}(z)p_{11}(u)}. \quad (3.3)$$

It is the transpose on both input factors here that supplies this reverse estimate.

Splitting the input vector. Choose a common orthonormal eigenbasis of the commuting A_0, A_1 . Let P be the projection onto the basis vectors on which the eigenvalue of A_0 is at least the eigenvalue of A_1 , and set

$$v = (P \otimes I_s)\chi, \quad w = ((I_r - P) \otimes I_s)\chi.$$

Then $\chi = v + w$. Because $PA_i(I_r - P) = 0$, the cross terms in the quadratic forms defining p_{ij} vanish. The eigenvalue comparison on the two subspaces, together with $B_0, B_1 \succeq 0$, therefore gives

$$\begin{aligned} p_{ij} &= p_{ij}(v) + p_{ij}(w), \\ p_{11}(v) &\leq p_{01}(v), \quad p_{00}(w) \leq p_{10}(w). \end{aligned} \quad (3.4)$$

For example, $A_0 - A_1$ is positive on $\text{ran } P$, so its tensor product with B_1 has nonnegative quadratic form on v ; the second comparison follows on $\ker P$ using B_0 .

We can now bound the four terms in the expansion of $C(\chi\chi^*)$. For the diagonal terms, (3.2) and (3.4) give

$$\begin{aligned} \|Z(v, v)\|_1 &\leq \sqrt{p_{00}(v)p_{11}(v)} \leq \sqrt{p_{00}p_{01}}, \\ \|Z(w, w)\|_1 &\leq \sqrt{p_{00}(w)p_{11}(w)} \leq \sqrt{p_{10}p_{11}}. \end{aligned}$$

For $Z(v, w)$ use the reverse estimate (3.3); for $Z(w, v)$ use the direct estimate (3.2). Both right sides become the same:

$$\begin{aligned} \|Z(v, w)\|_1 &\leq \sqrt{p_{00}(w)p_{11}(v)} \leq \sqrt{p_{10}p_{01}}, \\ \|Z(w, v)\|_1 &\leq \sqrt{p_{00}(w)p_{11}(v)} \leq \sqrt{p_{10}p_{01}}. \end{aligned}$$

Finally,

$$\sqrt{D_{00}}\omega\sqrt{D_{11}} = Z(v, v) + Z(v, w) + Z(w, v) + Z(w, w).$$

The triangle inequality now yields (3.1). $\square$

By Lemma 2.5, the same inequality applies to arbitrary joint positive matrices on all of Alice's and all of Bob's copies of a state in $\mathcal{C}$. Its lack of any normalization condition is the feature that will let us apply it at each complete public transcript.

4 Complete public transcripts

To apply the four-effect inequality to a protocol, we must obtain $X_i(t) \otimes Y_j(t)$ at each public transcript t , with the same two local pairs for all four bit values. These matrices describe the effects of the entire protocol on its input. They are not the effects of just its last measurements. We first derive this form for finite classical records, then prove its measurable counterpart for the admitted protocols below. The transcript always retains every public message, public random draw, and public stopping decision.

4.1 The finite-record mechanism

Let R be the input state on finite-dimensional spaces $A \otimes B$. In fixed product bases, Eve's canonical purification is

$$|\Psi_R\rangle = \sum_h \sqrt{R} |h\rangle_{AB} |h\rangle_E, \quad E \cong A \otimes B. \quad (4.1)$$

For the moment suppose there are finitely many local-instrument and public steps, including the final readouts, every classical record is finite, and each party produces its prescribed bit almost surely on R . Use the following resource rules: the recorded initial public variable C_0 is independent of the input and reference; conditional on C_0 , that input-reference state and the two private auxiliary resources form a product. The initial public law and local preparation kernels are fixed when the input varies. All later public-source draws use fixed input-independent kernels of the preceding public record. Thus the input is the only shared private resource.

Fix a complete public transcript t . In Alice's laboratory, compose the instrument branches consistent with t and final bit i , retaining private memory until it is no longer used and summing private outcomes locally. This gives a CP map $\mathcal{A}_i^t$ on her input. Bob obtains a CP map $\mathcal{B}_j^t$ in the same way. At this fixed public record each local choice can depend on the preceding public messages but not on the other party's unannounced private outcomes. The compatible private histories therefore sum independently within the two laboratories. Public-source weights are scalars fixed by t ; absorbing their product into either local map, the joint branch is $\mathcal{A}_i^t \otimes \mathcal{B}_j^t$.

Set $X_i(t) = (\mathcal{A}_i^t)^\dagger(I)$ and $Y_j(t) = (\mathcal{B}_j^t)^\dagger(I)$. These matrices are positive and the branch effect is $D_{ij}(t) = X_i(t) \otimes Y_j(t)$. Tracing the final laboratory memories out of that branch leaves Eve with

$$\tau_{ij}(t) = (\sqrt{R} D_{ij}(t) \sqrt{R})^\top.$$

Indeed, the (h, k) matrix entry on Eve's system is $\text{tr}[D_{ij}(t) \sqrt{R} |h\rangle \langle k| \sqrt{R}] = \langle k | \sqrt{R} D_{ij}(t) \sqrt{R} | h \rangle$. The same X_0, X_1 and Y_0, Y_1 occur in all four branches. Private instrument labels have been summed, not added to Eve's transcript. No pointwise normalization is required at a fixed t ; normalization holds after summing all final bits and all transcripts.

For a continuous transcript, a singleton may have probability zero, and with countably many steps an output may depend on the complete local history. The general statement will replace the preceding matrices by densities with respect to a single transcript measure. Its proof runs the protocol on a product of two local probes, each with an untouched reference. On that product input the local histories are independent conditional on the complete public record. Reference measurements recover common product matrix densities from this scalar independence, and a fixed filter on the references gives the blocks for R .

4.2 The admitted protocol model

The instrument formalism originates with Davies and Lewis [18]. Chitambar, Leung, Mančinska, Ozols, and Winter [14] distinguish finite-round LOCC, infinite linked protocols, and topological

closure. We specify the measurable model used here so that both its complete-history law and any terminal readout are part of the protocol data. Ordinary finite protocols obtain these laws by composition; a countable schedule requires the compatibility conditions below.

Definition 4.1 (Admitted local protocols). Let A and B be finite-dimensional complex input spaces. An admitted protocol consists of local quantum instruments, local memories and randomness, and authenticated two-way public classical communication. Each party may process its whole input space jointly. Its data and their joint laws satisfy the following requirements.

1. *Initial resources.* Before interaction, there is a recorded standard Borel public variable C_0 , possibly constant, whose law is fixed once the protocol is chosen. It is independent of the supplied input together with any purifying or untouched reference. Conditional on C_0 , the joint state or law of that input and reference and the two initial private auxiliary resources is the product of the input-reference state and one resource for each laboratory. These local resources include private ancillas and randomness and may depend on C_0 . The public law and these two preparation kernels are the same whenever this protocol is run on another input. All shared initial randomness is included in C_0 . An initialization acting on a party's input is treated as a first local operation. Apart from this recorded public variable and the supplied input, there is no shared resource and no preshared secret key.
2. *Local instruments.* An instrument assigns completely positive operations to measurable outcome sets, countably additively on input states. It has measurable outcome laws and measurable conditional local updates, and its total operation is trace preserving. Its classical outcome space is standard Borel. Local instruments may depend measurably on the local private history and the preceding public record.
3. *Local aggregation and chronological history.* A local subroutine completed before a later step may be aggregated into one instrument when that aggregate is supplied as a local instrument reproducing its retained output and reference law on arbitrary inputs with untouched finite-dimensional references. It includes every private record later used and its continuing memory. Its total is normalized, including a nonreturn or failure tag when needed. All public communication remains in the public schedule. After any such aggregation and before a separate terminal readout on a complete infinite history, the genuine classical history, including private instrument outcomes and public records, forms a finite sequence or a sequence indexed by $\mathbb{N}$, in fixed standard Borel coordinate spaces. Instrument-choice tags and fixed padding for stopped histories are included in these spaces. A local choice uses only that laboratory's preceding private history and the preceding public record; any choice affecting the other laboratory is communicated through a public step.
4. *Public steps and the complete record.* The public discussion is correspondingly finite or indexed by $\mathbb{N}$. Each message is produced by its sender's local process from that sender's private history and the preceding public record. For a later public-source draw after the public record $T_m = t$, the recorded outcome is drawn from a measurable probability kernel $\kappa_m(t, dc)$ depending only on that record. On every unnormalized joint instrument law, this public update multiplies by the scalar kernel and acts as the identity on the quantum and reference registers. The kernel at a given record is the same for every initial input. A party-generated announced choice, deterministic or randomized, is that party's message. Public choices of speaker and public stopping decisions use these same two rules; a choice measurable from the preceding public record is a Dirac public kernel. Simultaneous messages may be ordered while retaining their dependence on the earlier record. Every publicly observed control or other datum is generated

and recorded by one of these steps. The standard Borel record T is the encoded sequence of C_0 and all these public messages, draws, speaker choices, and stopping decisions. Private outcomes and quantum memories remain in their own laboratories.

5. *Reference-compatible prefix laws.* The specified steps are one family of instruments on arbitrary positive trace-class inputs, with their amplifications by any untouched finite-dimensional reference. Their unnormalized branch laws give countably additive positive matrix-valued measures on that reference for finite prefixes. Integrating the next record recovers the preceding reference measure after the current laboratory memories are traced out.

6. *Terminal outputs and local quantum readouts.* A terminal classical output may be a measurable function of that laboratory's instrument history and the received complete transcript T . A decision to invoke a terminal quantum readout must be measurable in the same local records.

For a terminal quantum readout, the protocol supplies both its incoming local memory and the instrument applied to it. The memory is supplied by one fixed measurable local CP process on those histories, defined on arbitrary inputs with any untouched finite-dimensional reference. The readout is a specified measurable local instrument on that memory, selected using only that laboratory's private history and T . The two laboratories refine the reference measure on the complete history by these local operations, acting as the identity on the reference.

Together with the preceding schedule, these refinements give one countably additive, unnormalized instrument law on the complete history and the terminal outcomes, for all initial inputs and finite untouched references. Include a tag for a missing or noninvoked readout among each laboratory's outcomes. Summing either laboratory's terminal outcomes removes that local refinement. Summing both and tracing the output memories recovers the reference measure on the complete history with its stated finite-prefix marginals.

After any terminal outcomes have been included, each prescribed output status is a measurable local value in the finite output alphabet together with a symbol $\perp$ for nonproduction. An output already produced is retained. The incoming-memory and compatibility requirements concern only memory used by a terminal readout; unused memory need not converge. An unspecified evolution through infinitely many quantum operations does not by itself define a terminal readout.

7. *Completion on a specified input.* A *completed bit-output protocol on a specified input* has a bit in each laboratory almost surely on that input, with normalized joint output. Almost-sure production is required only on this actual input. More generally a completed key-output protocol produces an output in a specified finite classical alphabet at each party almost surely on the actual input, again with normalized joint output. Trace-norm limits of completed actual outputs, with full recoverable public information, will be treated separately in Section 5.

For a finite protocol whose instruments and public steps satisfy clauses 1–4, composition of those instruments supplies the prefix laws and any finite terminal refinement in clauses 5–6. The separate complete-history and terminal-process requirements matter when a countable schedule or an aggregated subroutine is used. There is no uniform bound on the amount of communication, the size of a standard Borel message, or the number of rounds within this scope. In particular, a terminal classical bit can depend on an entire infinite record.

4.3 Common effects at a complete transcript

Lemma 4.2 (Common local factors for a complete transcript). *Let R be a density operator on finite-dimensional complex spaces $A \otimes B$, and let an admitted protocol produce bits $i, j \in \{0, 1\}$ almost surely on R . Give Eve the canonical purification (4.1), and let her retain E and the complete public transcript T with value t in its standard Borel space $\mathbb{T}$. There exist a finite positive measure μ on $\mathbb{T}$ and measurable positive matrices $X_0(t), X_1(t)$ on A and $Y_0(t), Y_1(t)$ on B such that the positive matrix-valued measure of Eve in bit branch (i, j) has the density*

$$\tau_{ij}(t) = (\sqrt{R}[X_i(t) \otimes Y_j(t)]\sqrt{R})^\top \quad \text{with respect to } \mu. \quad (4.2)$$

The same $X_i(t)$ and $Y_j(t)$ work for all four branches outside one μ -null set, and

$$\sum_{i,j=0}^1 \int_{\mathbb{T}} \text{tr } \tau_{ij}(t) d\mu(t) = 1.$$

The transpose in (4.2) is in the product basis identifying E with $A \otimes B$. No invertibility of R is assumed.

Proof. Write $a = \dim A$ and $b = \dim B$, and attach untouched finite references $A_0 \cong A$ and $B_0 \cong B$. Run the protocol on the product of the normalized local probes

$$|\Phi_a\rangle_{AA_0} \otimes |\Phi_b\rangle_{BB_0}, \quad |\Phi_a\rangle = a^{-1/2} \sum_{r=1}^a |r, r\rangle, \quad |\Phi_b\rangle = b^{-1/2} \sum_{s=1}^b |s, s\rangle. \quad (4.3)$$

The protocol need not produce both bits on this auxiliary input. For this experiment assign bit zero in a laboratory if its prescribed bit is never produced, and retain a bit that has already been produced. The assignment is a measurable local rule. It does not replace the other laboratory's output when only one output is missing. We will remove the contribution of these auxiliary defaults after recovering the actual input.

The proof has three steps. First, clauses 1–4 give conditional independence of the two local sampling tapes on the product probes, given the complete public transcript. Next, clauses 5–6 give positive reference measures; informationally complete tests turn the scalar independence into product matrix densities. Finally, a filter on the untouched references prepares (4.1) and transfers those densities to Eve's actual blocks.

Conditional independence on the product probes. The scalar factorization has the finite private-coin rectangle property as a classical antecedent [6, Lemma 6.7]. We prove the complete-history version needed here before recovering its matrix-valued factors.

Choose a finite informationally complete POVM on each reference. One explicit construction in dimension d starts with the projections onto the basis vectors and onto

$$(|r\rangle + |s\rangle)/\sqrt{2}, \quad (|r\rangle + i|s\rangle)/\sqrt{2} \quad (r < s).$$

These projections span the real vector space of Hermitian matrices. If they are $P_1, \dots, P_N$, the $2N$ effects P_k/N and $(I - P_k)/N$ form a POVM with the same spanning property. Call the two resulting POVMs $(E_r)_r$ and $(G_s)_s$. Because the references are untouched, these tests commute with all laboratory operations. Their joint statistics with the transcript and final bits can therefore be computed with the tests performed at the start.

In this tested product-probe experiment, sample C_0 with the same fixed law as in the actual run. Conditional on C_0 , all local outcomes can be sampled using independent local random tapes U for Alice and V for Bob. The tapes may be taken as countable products of uniform variables, with coordinates reserved for the reference tests and any terminal readouts. They are chosen before the sampling; the local outcome histories are then functions of these tapes and the public record. At each local instrument, sample its conditional outcome law using a fresh uniform variable on that laboratory's tape and apply the stipulated conditional state update. The tested input and private ancillas are a product across laboratories conditional on C_0 , and each local update changes only its own local state. At a public-source step, sample $\kappa_m(T_m, dc)$ using a fresh public uniform variable independent of the private tapes and all prior sampling variables; only the recorded draw enters the transcript. This step leaves the quantum registers unchanged. Induction over the specified finite prefixes therefore shows that this sampling has the genuine joint finite-prefix laws. For a standard Borel outcome space a probability kernel has such a jointly measurable sampler: one may code the space by a Borel subset of $[0, 1]$ and use the measurable kernel-sampling theorem [36, Appendix A1, Theorem A1.2, and Lemma 3.22]. The initial reference tests and each specified local terminal quantum readout are sampled in the same way. Their stated local compatible refinements give the genuine tested laws on the complete histories. A terminal classical readout is already a measurable function of its records. Each defaulted bit and each reference-test result is consequently a measurable function of that laboratory's tape and the complete public transcript. This sampling describes the auxiliary product input; the actual input R need not have a product simulation.

Let T_m consist of C_0 and the first m subsequent public records, and let $\mathcal{F}_m = \sigma(T_m)$. At a step sent by Alice the new record is measurable with respect to $\sigma(U) \vee \mathcal{F}_m$; at a step sent by Bob the analogous statement holds with V . At a public-source step with new draw C , the sampling construction gives

$$\mathbb{P}(C \in dc \mid \sigma(U, V) \vee \mathcal{F}_m) = \kappa_m(T_m, dc).$$

We show by induction that U and V are conditionally independent given $\mathcal{F}_m$. The assertion at $m = 0$ follows from the product initial resources and independent private tapes conditional on C_0 .

Suppose it holds at m and Alice sends the next record. For bounded measurable functions u and v , conditional independence gives

$$\mathbb{E}[v(V) \mid \sigma(U) \vee \mathcal{F}_m] = \mathbb{E}[v(V) \mid \mathcal{F}_m].$$

As $\mathcal{F}_{m+1} \subseteq \sigma(U) \vee \mathcal{F}_m$, conditioning this equality on $\mathcal{F}_{m+1}$ shows that

$$\mathbb{E}[v(V) \mid \mathcal{F}_{m+1}] = \mathbb{E}[v(V) \mid \mathcal{F}_m].$$

Using the same equality inside the conditional expectation of $u(U)v(V)$ gives

$$\mathbb{E}[u(U)v(V) \mid \mathcal{F}_{m+1}] = \mathbb{E}[u(U) \mid \mathcal{F}_{m+1}]\mathbb{E}[v(V) \mid \mathcal{F}_{m+1}]. \quad (4.4)$$

The Bob step is symmetric.

For a public-source step, let φ be a bounded measurable function of its new draw C , and write $(\kappa_m \varphi)(T_m) = \int \varphi(c) \kappa_m(T_m, dc)$. The kernel condition and the induction hypothesis give

$$\mathbb{E}[u(U)v(V)\varphi(C) \mid \mathcal{F}_m] = (\kappa_m \varphi)(T_m) \mathbb{E}[u(U) \mid \mathcal{F}_m] \mathbb{E}[v(V) \mid \mathcal{F}_m].$$

Testing against bounded $\mathcal{F}_m$ -measurable factors times $\varphi(C)$ and applying the monotone class theorem identifies conditional expectations on $\mathcal{F}_{m+1} = \mathcal{F}_m \vee \sigma(C)$. Taking $v = 1$ or $u = 1$ shows that the respective conditional marginal is unchanged; taking both functions gives their product.

Thus the same factorization (4.4) holds after the public-source step. The rules for speaker and stopping records are among these sender and public-source steps. If two messages were simultaneous, ordering them preserves the required measurability for the second, because it still depends only on its original earlier record. Fixed padding after a public stop also preserves the claim.

Set $\mathcal{F}_\infty = \sigma(\bigcup_m \mathcal{F}_m) = \sigma(T)$, the sigma field of the actual complete transcript. The auxiliary sampling uniforms for later public draws are not adjoined. If Z is bounded, then

$$\mathbb{E}[Z \mid \mathcal{F}_m] \longrightarrow \mathbb{E}[Z \mid \mathcal{F}_\infty] \quad \text{in } L^2.$$

Indeed, these are the orthogonal projections of Z onto increasing closed subspaces of L^2 . Their limit is the projection onto the closure of the union. Indicators of finite-prefix events belong to that union, and the monotone class theorem shows that their span is dense in $L^2(\mathcal{F}_\infty)$. The limiting projection is therefore the indicated conditional expectation. For bounded u, v , the two conditional expectations on the right of (4.4) converge in L^2 and remain bounded. Cauchy–Schwarz applied to the difference of their products gives convergence in L^1 . Passing to the limit in (4.4) proves

$$\mathbb{E}[u(U)v(V) \mid \mathcal{F}_\infty] = \mathbb{E}[u(U) \mid \mathcal{F}_\infty]\mathbb{E}[v(V) \mid \mathcal{F}_\infty].$$

The tape spaces and $\mathbb{T}$ are standard Borel. They admit regular conditional probability kernels and conditional integration of functions depending on t [36, Theorems 6.3–6.4]. Apply the last factorization to indicators from countable generating algebras for the two tape spaces, including the whole spaces, and discard the union of the resulting null sets. Their product rectangles form a π -system generating the product sigma field. Uniqueness of probability measures on that π -system then yields, on a single set of full transcript measure,

$$P_{U,V|T=t} = P_{U|T=t} \otimes P_{V|T=t}. \quad (4.5)$$

In particular this law factors the conditional expectations of any bounded measurable function of (U, t) and any bounded measurable function of (V, t) . Thus the pair consisting of Alice’s defaulted bit and her reference-test result is conditionally independent of the corresponding pair for Bob, given t .

Positive reference measures and their densities. Return to the experiment in which the probes are retained and the reference tests are not performed. We first construct its reference matrix measure on the space of all genuine classical instrument histories, including private histories. The fixed initial law of C_0 and the specified finite-prefix instruments give a positive reference matrix measure at every prefix. Summing the next local outcome recovers the preceding reference measure because the total instrument is trace preserving. Integrating a public-source outcome does the same because its scalar kernel has total mass one and acts as the identity on the quantum registers. These measures are therefore consistent on finite cylinders.

Here is the countable extension in finite-dimensional terms. For a positive reference test, its scalar finite-prefix laws are consistent finite measures; after normalization, or by adjoining the complementary test outcome, the countable product extension theorem applies [36, Theorem 6.14]. It supplies a finite scalar measure on the countable standard Borel history space. Recover matrix entries from finitely many rank-one reference tests by polarization. For an arbitrary reference vector z , extend also its positive rank-one test. On finite cylinders the resulting scalar measure equals the quadratic form $z^*W(\cdot)z$ of the recovered matrix measure. Uniqueness of finite measures on cylinders extends this equality to the entire history sigma field. Thus $z^*W(S)z \geq 0$ for every measurable S and every z , so W is a positive matrix-valued measure. This argument constructs a measure for the untouched finite references; it asks for no limiting state of unused quantum memory.

Measurable terminal classical readouts and the local default rule now give measurable events in the history space. If a terminal quantum readout is used, the incoming local process stipulated in Definition 4.1, on those same instrument histories and with an untouched finite reference, together with its specified local instrument gives the bit-conditioned positive reference measures. Each instrument is selected from its own laboratory's history and T ; their joint refinement is obtained by the two local operations. Include the missing or noninvoked tag among the outcomes, so summing both local outcomes recovers the reference measure on the complete history. This terminal operation is a separate extension of that history; its outcome is included in the enlarged history before the classical readout. This is where clauses 5–6 supply the needed locality, prefix compatibility, and reference stability. Write $\widehat{W}$ for the resulting positive reference measure on the enlarged history space. Its marginal on the preterminal history is W ; when no readout is used, identify $\widehat{W}$ with W using trivial terminal tags. The construction uses the genuine instrument records; the tapes above were only a sampling device for their tested scalar laws.

Let $W_{ij}(S)$ be the restriction and pushforward of $\widehat{W}$ to the defaulted bit pair (i, j) and transcript in a measurable set $S \subseteq T$. Define the probe transcript law

$$\mu(S) = \sum_{i,j=0}^1 \text{tr } W_{ij}(S).$$

It is a probability measure: the probe is normalized, the instrument totals preserve trace, and the defaults assign a pair on every history. For each i, j , positivity and $\text{tr } W_{ij}(S) \leq \mu(S)$ imply absolute continuity of every matrix entry with respect to μ . Hence there are entrywise Radon–Nikodym densities $W_{ij}(t)$. Only the dominated finite-measure case is needed here. For completeness, if $0 \leq \nu \leq c\mu$, the functional $g \mapsto \int g d\nu$ on simple functions is bounded in $L^2(\mu)$ by Cauchy–Schwarz. Its Hilbert space representing vector is a density, and indicator tests show that the density lies between 0 and c . Apply this fact to the finitely many polarization tests for the matrix entries.

Testing the densities on a countable dense set of reference vectors shows that $W_{ij}(t) \succeq 0$ outside one null set. The equality defining μ also gives

$$\sum_{i,j} \text{tr } W_{ij}(t) = 1 \quad \text{for } \mu\text{-almost every } t.$$

Put

$$U_i(t) = \sum_j \text{tr}_{B_0} W_{ij}(t), \quad V_j(t) = \sum_i \text{tr}_{A_0} W_{ij}(t).$$

Both local pairs are positive, and $\sum_i \text{tr } U_i(t) = \sum_j \text{tr } V_j(t) = 1$ almost everywhere.

The conditional scalar statistics of the reference tests are the traces against these densities. Their local marginals are $\text{tr}[E_r U_i(t)]$ and $\text{tr}[G_s V_j(t)]$, because the other POVM sums to the identity. Conditional independence from (4.5) therefore gives

$$\text{tr}[(E_r \otimes G_s)W_{ij}(t)] = \text{tr}[E_r U_i(t)] \text{tr}[G_s V_j(t)]$$

for every i, j, r, s , outside one null set. The tensor products $E_r \otimes G_s$ span the Hermitian matrices on $A_0 \otimes B_0$. Equality against that spanning family proves

$$W_{ij}(t) = U_i(t) \otimes V_j(t) \quad \text{for all } i, j \text{ and } \mu\text{-almost every } t. \quad (4.6)$$

We have obtained the same two local matrix pairs for all four branches, without conditioning on a positive-probability singleton transcript and without making private instrument labels public.

Recovering the actual purification. Identify $A_0 \otimes B_0$ with E in the product basis and define

$$K = \sqrt{ab}(\sqrt{R})^\top.$$

The elementary identity $(I \otimes S^\top) \sum_h |h, h\rangle = (S \otimes I) \sum_h |h, h\rangle$ shows that applying $I_{AB} \otimes K$ to the product probes (4.3) gives exactly $|\Psi_R\rangle$. This joint filter on the references is an algebraic preparation of the input purification; the laboratories are not required to implement it. It need not be trace preserving or a contraction.

The same independent law of C_0 and the same C_0 -conditioned local auxiliary resources are used on the probes and on the actual input. Consequently the filter also prepares the full initial joint law, including C_0 and those resources. On each finite cylinder of genuine histories, the reference matrix for this filtered input is $KW(\cdot)K^*$, because the references are untouched and the filter commutes with every local instrument. Every public-source update also commutes with the filter: at a given public record it multiplies both laws by the same scalar kernel and acts as the identity on the reference. The scalar entries on both sides are finite measures, so uniqueness extends their equality from cylinders to the full history sigma field. The same commutation holds for any appended terminal instrument by clause 6. Thus the filtered law on the enlarged history is $K\widehat{W}(\cdot)K^*$, and the identity holds for the bit-conditioned measures as well.

Let N be the event in this enlarged history space that at least one prescribed local bit is missing before the default rule. The event may depend on private records. On the actual input its probability is zero by hypothesis, so positivity gives

$$\text{tr}[K\widehat{W}(N)K^*] = 0, \quad K\widehat{W}(N)K^* = 0.$$

Every subevent of N has a positive filtered matrix bounded by this zero matrix. It therefore also contributes zero. Thus the local defaults leave the actual output measure unchanged, even though they may occur on the probes. The argument was made on full histories before discarding private records, so N need not be public.

It follows from (4.6) that the actual Eve density in branch (i, j) is $K(U_i(t) \otimes V_j(t))K^*$. The matrix $(\sqrt{R})^\top$ is Hermitian. Hence direct multiplication gives

$$\begin{aligned} K(U_i \otimes V_j)K^* &= ab(\sqrt{R})^\top(U_i \otimes V_j)(\sqrt{R})^\top \\ &= (\sqrt{R}[aU_i^\top \otimes bV_j^\top]\sqrt{R})^\top. \end{aligned}$$

Take $X_i(t) = aU_i(t)^\top$ and $Y_j(t) = bV_j(t)^\top$, redefining them on a null set if necessary. Transposition preserves positivity, so these are measurable positive local matrices. The filtered total trace is one because R is normalized, and (4.2) follows. No inverse of R appears anywhere in the construction. $\square$

The canonical purification is convenient for the formula, but it does not restrict the adversary or the ideal comparison law. The following observation will let the secret-bit proof use that formula for every purifying space.

Lemma 4.3 (Recovery from another purifying space). *Let R and $|\Psi_R\rangle$ be as in Lemma 4.2, and let $|\psi\rangle \in A \otimes B \otimes \tilde{E}$ be any purification of R . There is a channel $\mathcal{R}$ from $\tilde{E}$ to E that recovers the canonical purification. If Alice and Bob run the same admitted protocol and Eve retains her purification and the full transcript, $\mathcal{R}$ acting on the purifying register and the identity acting on the transcript recover the canonical output of Lemma 4.2. Moreover this channel maps every ideal secret-bit law with Eve in $\tilde{E}T$ to an ideal secret-bit law with Eve in ET .*

Proof. Let $\mathcal{S} = \text{supp}(R^\top) \subseteq E$. Schmidt decomposition gives an isometry $V : \mathcal{S} \rightarrow \tilde{E}$ with $|\psi\rangle = (I_{AB} \otimes V)|\Psi_R\rangle$; the canonical vector is supported on $\mathcal{S}$ in its purifying factor. Choose any density operator ζ on E , and define, with the compression term included into E from $\mathcal{S}$,

$$\mathcal{R}(\xi) = V^* \xi V + \text{tr}[(I_{\tilde{E}} - VV^*)\xi] \zeta.$$

Compression is CP, as is the positive trace functional followed by preparation of ζ . Their sum preserves trace, so $\mathcal{R}$ is a channel. Its first term inverts V on the actual purifying support, and the second term vanishes there. Local operations on A, B keep each conditional purifying state supported in $\text{ran } V$, and public-source steps act as the identity on that register, so recovery is exact in every transcript and bit branch. The same formula is valid when $\tilde{E}$ is larger than the finite purifying support, acting on trace-class inputs.

An ideal bit has the form $\frac{1}{2} \sum_{i=0}^1 |i, i\rangle \langle i, i| \otimes \sigma$. Applying $\mathcal{R} \otimes \text{id}_T$ to its Eve factor replaces σ by another normalized state and leaves that form intact. This includes ideal states with mass outside $\text{ran } V$. Thus trace-norm contraction transfers any lower bound proved for the canonical output and every canonical ideal law to the output for the arbitrary purification and every ideal law in its larger space. $\square$

5 Separation from an ideal secret bit

We now combine the common factors at a complete transcript with the four-effect inequality. A nearly correct bit has little weight in the 01 and 10 branches. A nearly uniform secret bit requires Eve's 00 and 11 blocks to be close and to carry nearly half the total weight each. Root fidelity makes these two requirements comparable, and the resulting estimates give a uniform numerical gap.

Throughout the paper the trace norm is unhalved:

$$\|H\|_1 = \text{tr} \sqrt{H^* H}.$$

For a classical standard Borel register with a finite-dimensional quantum register, a state is a positive matrix-valued measure of total trace one. If a Hermitian difference of such measures has density $H(t)$ with respect to a common finite dominating measure λ , its trace norm is $\int \|H(t)\|_1 d\lambda(t)$. This value is independent of the chosen dominating measure by homogeneity. The usual trace distance between normalized states is one half of this norm.

We use trace-norm contraction on Hermitian differences under channels, including classical readouts. In finite dimensions it follows directly from duality: the adjoint of a trace-preserving positive map is unital and positive, so it sends each Hermitian contraction Q , with $-I \preceq Q \preceq I$, to another Hermitian contraction. Taking the supremum of $\text{tr}(QH)$ over such Q proves contraction. The same argument applies to trace-class quantum registers and classical matrix-valued measures through their bounded observables.

Definition 5.1 (Distillable secret key). Let ω be a bipartite state, with Eve initially holding a purification. A key-distillation sequence uses an admitted protocol of Definition 4.1 on each $\omega^{\otimes n}$, allowing each party to process all its n copies jointly. The normalized classical output at each party is a string of a fixed length $\ell_n \in \mathbb{Z}_{\geq 0}$ for that n . Eve's output register E' retains the initial purification and the complete public transcript. For any normalized state or classical-quantum law $\sigma_{E'}$, define the ideal length- ℓ key by

$$\gamma_\sigma^{(\ell)} = 2^{-\ell} \sum_{z \in \{0,1\}^\ell} |z, z\rangle \langle z, z| \otimes \sigma_{E'}.$$

The sequence is secure if there are normalized comparison laws $\sigma_{E'}^{(n)}$ such that

$$\left\| \tau_{K_A K_B E'}^{(n)} - \gamma_{\sigma^{(n)}}^{(\ell_n)} \right\|_1 \longrightarrow 0. \quad (5.1)$$

Its asymptotic lower rate is $\liminf_{n \rightarrow \infty} \ell_n/n$. The distillable secret key $K_D(\omega)$ is the supremum of these rates over secure sequences, including the zero rate obtained by producing the empty key. A positive rate therefore requires $\ell_n \geq 1$ for all sufficiently large n .

For positive matrices define the unnormalized root fidelity by

$$f(H, J) = \left\| \sqrt{H} \sqrt{J} \right\|_1.$$

In particular $f(cH, cJ) = cf(H, J)$ for $c \geq 0$. The next facts hold at singular matrices as well as invertible ones. The lower comparison is the unnormalized form associated with the Powers–Størmer inequality [50, Lemma 4.1]; for normalized states it is the lower Fuchs–van de Graaf bound [24, Theorem 1]. We give the short matrix proof needed at our exact normalization.

Lemma 5.2 (Root fidelity and trace norm). *For arbitrary square complex matrices A, B of the same size,*

$$f(A^*A, B^*B) = \|AB^*\|_1. \quad (5.2)$$

For positive matrices H, J of the same size,

$$f(H, J) \geq \frac{\text{tr } H + \text{tr } J - \|H - J\|_1}{2}. \quad (5.3)$$

Also $f(H^\top, J^\top) = f(H, J)$.

Proof. Extend the partial isometries in the polar decompositions of the square matrices to unitaries, writing $A = U\sqrt{A^*A}$ and $B = V\sqrt{B^*B}$. Then $AB^* = U\sqrt{A^*A}\sqrt{B^*B}V^*$, and unitary invariance of the trace norm proves (5.2). This extension is available even when either matrix is singular.

For (5.3), put $A = \sqrt{H}$, $B = \sqrt{J}$, and $Z = A - B$. The matrix Z is Hermitian and $H - J = ZA + BZ$. The spectral sign $\text{sgn}(Z)$ is a Hermitian contraction. Trace-norm duality and cyclicity therefore give

$$\|H - J\|_1 \geq \text{tr}[\text{sgn}(Z)(H - J)] = \text{tr}[|Z|(A + B)].$$

In a unit eigenvector of Z with eigenvalue z , positivity of A, B implies

$$\langle A + B \rangle \geq |\langle A - B \rangle| = |z|.$$

Summing in an eigenbasis of Z shows $\text{tr}[|Z|(A + B)] \geq \text{tr } Z^2$. Finally,

$$\text{tr } Z^2 = \text{tr } H + \text{tr } J - 2\text{tr}(AB), \quad 0 \leq \text{tr}(AB) \leq \|AB\|_1 = f(H, J).$$

Combining these inequalities proves the lower bound. No commutation of H and J was used. For the transpose identity, functional calculus gives $\sqrt{H}^\top = (\sqrt{H})^\top$. The product of the two transposed square roots is $(\sqrt{J}\sqrt{H})^\top$; transpose and adjoint preserve trace norm, proving $f(H^\top, J^\top) = f(H, J)$. $\square$

Theorem 5.3 (Uniform obstruction to a secret bit). *Let $\omega \in \mathcal{C}$. For every integer $n \geq 1$, let an admitted completed protocol on $\omega^{\otimes n}$ produce normalized bit registers K_A, K_B , and let its output*

$\tau_{K_A K_B E'}$ retain in E' an initial purification held by Eve and the complete public transcript. The purification may be arbitrary. For every normalized state or classical-quantum law $\sigma_{E'}$,

$$\left\| \tau_{K_A K_B E'} - \frac{1}{2} \sum_{i=0}^1 |i, i\rangle \langle i, i| \otimes \sigma_{E'} \right\|_1 \geq \frac{1}{5}. \quad (5.4)$$

Thus the conventional trace distance is at least $1/10$, and $K_D(\omega) = 0$ in Definition 5.1.

Proof. Group all of Alice's factors and all of Bob's factors, and set $R = \omega^{\otimes n}$. Lemma 2.5 gives $R \in \mathcal{C}$. First apply the classical readout of Eve's retained complete public transcript, leaving her initial purifying register unchanged, and discard any extra retained registers. On the actual output this produces the full classical transcript together with the purification. On any ideal bit comparison state it produces an ideal bit with a classical-quantum comparison law. Next, Lemma 4.3 supplies a channel from the retained purifying space to the canonical one, leaving the transcript unchanged. It recovers the actual canonical output and again sends every ideal law to an ideal law. Trace-norm contraction under these Eve-side channels therefore reduces the claim to the canonical purification (4.1) and the full classical transcript T .

Apply Lemma 4.2 to obtain a finite transcript measure μ , the positive factors $X_i(t), Y_j(t)$, and the Eve densities $\tau_{ij}(t)$ of (4.2). Fix any normalized positive matrix-valued comparison law σ_{ET} on the same transcript space.

Let ν be the transcript law of σ_{ET} , namely $\nu(S) = \text{tr } \sigma_{ET}(S)$, and use the common finite measure $\lambda = \mu + \nu$. If $h = d\mu/d\lambda$, replace each actual density by $h(t)\tau_{ij}(t)$ and both $X_i(t)$ by $h(t)X_i(t)$, leaving the $Y_j(t)$ unchanged. Equation (4.2) then still holds, including where $h = 0$. The ideal law has a positive density $\sigma(t)$ with respect to λ and $\int \text{tr } \sigma(t) d\lambda(t) = 1$. Homogeneity preserves all actual probabilities and root-fidelity integrals in this change of measure. We now denote λ again by μ . This common measure keeps singular parts of either original transcript law; there is no assumption that the ideal and actual laws have the same support.

For these common densities set

$$D_{ij}(t) = X_i(t) \otimes Y_j(t), \quad p_{ij}(t) = \text{tr } \tau_{ij}(t) = \text{tr}[D_{ij}(t)R], \quad P_{ij} = \int p_{ij}(t) d\mu(t).$$

All P_{ij} are nonnegative and $\sum_{i,j} P_{ij} = 1$. With $A = \sqrt{D_{00}(t)}\sqrt{R}$ and $B = \sqrt{D_{11}(t)}\sqrt{R}$, the transpose invariance and (5.2) give

$$\begin{aligned} f(\tau_{00}(t), \tau_{11}(t)) &= f(\sqrt{R}D_{00}(t)\sqrt{R}, \sqrt{R}D_{11}(t)\sqrt{R}) \\ &= \left\| \sqrt{D_{00}(t)}R\sqrt{D_{11}(t)} \right\|_1. \end{aligned} \quad (5.5)$$

The privacy interpretation of an off-diagonal coherence norm as the fidelity of conditional Eve states is also used in the private-state approach [30, Eq. (7)]. Here the identity is applied to unnormalized branches at each complete transcript. Its proof includes singular R and singular effects.

Apply Lemma 3.2 at almost every t . Scalar Cauchy–Schwarz after integration gives

$$\begin{aligned} \mathfrak{f} &:= \int f(\tau_{00}(t), \tau_{11}(t)) d\mu(t) \\ &\leq \sqrt{P_{00}P_{01}} + \sqrt{P_{10}P_{11}} + 2\sqrt{P_{01}P_{10}}. \end{aligned} \quad (5.6)$$

For example, $\int \sqrt{p_{00}(t)p_{01}(t)} d\mu(t)$ is at most $\sqrt{P_{00}P_{01}}$. The factors need only be positive; no pointwise POVM normalization was used.

We next express what small distance from the fixed ideal bit would require of this same $\mathfrak{f}$. Let η be the unhalved trace-norm distance after the common transcript readout, and define

$$\begin{aligned} e &= P_{01} + P_{10}, \\ q &= \int \|\tau_{00}(t) - \tau_{11}(t)\|_1 d\mu(t), \\ a_i &= \int \left\| \tau_{ii}(t) - \frac{1}{2}\sigma(t) \right\|_1 d\mu(t) \quad (i = 0, 1). \end{aligned}$$

The unequal-bit blocks of the ideal state are zero. Additivity of the trace norm across the classical bit blocks, and then the triangle inequality within the two equal-bit blocks, give

$$\eta = e + a_0 + a_1, \quad q \leq a_0 + a_1 = \eta - e. \quad (5.7)$$

Trace conservation gives one further inequality that is needed for the uniform constant. The actual equal-bit blocks have total trace $1 - e$, whereas the corresponding ideal blocks have total trace one. Thus

$$a_0 + a_1 \geq \left| \sum_{i=0}^1 \int \text{tr}(\tau_{ii}(t) - \frac{1}{2}\sigma(t)) d\mu(t) \right| = e,$$

and (5.7) implies

$$e \leq \frac{\eta}{2}. \quad (5.8)$$

Integrating (5.3) gives

$$\mathfrak{f} \geq \frac{P_{00} + P_{11} - q}{2} = \frac{1 - e - q}{2} \geq \frac{1 - \eta}{2}.$$

On the other hand, Cauchy–Schwarz on the first two terms of (5.6) and the arithmetic–geometric mean inequality on the last give

$$\begin{aligned} \sqrt{P_{00}P_{01}} + \sqrt{P_{10}P_{11}} &\leq \sqrt{(P_{00} + P_{11})(P_{01} + P_{10})} = \sqrt{e(1 - e)}, \\ 2\sqrt{P_{01}P_{10}} &\leq e. \end{aligned}$$

The two estimates have the exact combined form

$$\frac{1 - \eta}{2} \leq \mathfrak{f} \leq \sqrt{e(1 - e)} + e, \quad e \leq \frac{\eta}{2}. \quad (5.9)$$

If $\eta < 1/5$, then $e < 1/10$. On $0 \leq e \leq 1/10$ the function $\sqrt{e(1 - e)} + e$ is increasing and has value $2/5$ at $e = 1/10$. The right side of (5.9) is therefore strictly less than $2/5$, while its left side is strictly greater than $2/5$. This contradiction proves (5.4), uniformly in n , the represented state, the protocol, and the ideal comparison law.

Finally suppose a secure key-distillation sequence had $\ell_n \geq 1$ for infinitely many n . Along those indices, the local classical channels that retain the first bit of each string and discard the other bits, while retaining E' , map $\gamma_{\sigma(n)}^{(\ell_n)}$ to an ideal bit: exactly half the uniform strings have either first bit. Trace-norm contraction would make the resulting bit-output error tend to zero, contradicting (5.4). Thus every secure sequence has $\ell_n = 0$ for all sufficiently large n . In particular its asymptotic lower rate is zero. The empty key achieves rate zero, so $K_D(\omega) = 0$. $\square$

The proof also yields the retained conditioning and limiting forms of the obstruction. These statements specify how normalization and Eve’s complete information are preserved.

Proposition 5.4 (Conditioning, aborts, and completed-output limits). *Consider inputs $\omega^{\otimes n}$ with $\omega \in \mathcal{C}$ and $n \geq 1$, and admitted protocols whose normalized completed outputs retain in Eve's register an arbitrary initial purification and the complete public transcript. The following statements extend the gap in Theorem 5.3; in item 2, the completed local output alphabet may include an abort symbol.*

1. *If A is a measurable event of the complete public transcript with probability $r > 0$, the normalized bit output conditioned on A has distance at least $1/5$ from every ideal bit law for that accepted output. This holds for every $r > 0$.*
2. *Suppose a sequence of completed protocols has a locally available abort symbol at either party. If the probability that at least one party aborts tends to zero, and the normalized non-aborting outputs approach ideal non-aborting bit laws, then replacing each local abort by the fixed bit zero would give completed bit outputs approaching ideal bits. Such a sequence is impossible for inputs in $\mathcal{C}$ (including any numbers of copies).*
3. *Let τ_k be completed normalized bit outputs to which the gap applies, possibly on different output spaces. Suppose their bit registers are kept fixed and their Eve registers are embedded in a common output space F , giving states $\hat{\tau}_k$. The embeddings must preserve and permit recovery of the full initial purification and complete public records, without coarse-graining those records. In particular, an Eve-side recovery channel recovers τ_k from $\hat{\tau}_k$ while leaving the bits fixed. If $\hat{\tau}_k \rightarrow \tau$ in trace norm, then for every normalized law σ_F ,*

$$\left\| \tau - \frac{1}{2} \sum_{i=0}^1 |i, i\rangle \langle i, i| \otimes \sigma_F \right\|_1 \geq \frac{1}{5}.$$

Proof. For the first assertion, use the canonical factors from Lemma 4.2. Restrict their measure μ to A and replace both $X_0(t), X_1(t)$ by $X_0(t)/r, X_1(t)/r$, leaving the $Y_j(t)$ unchanged. Each of the four blocks in (4.2) is divided by r , and its integral over A is the corresponding normalized accepted block. All probabilities and the root fidelity scale by the same factor $1/r$. The accepted total trace is one, so the proof of (5.9), including $e \leq \eta/2$, applies without change. The recovery and common-measure arguments cover arbitrary purifications and arbitrary ideal laws for the accepted output. No positive lower bound on r was used.

For the second assertion, let δ_k be the probability of at least one abort, let τ_k^{ok} be the normalized non-aborting state when $\delta_k < 1$, and let ξ_k be the normalized state on the aborting part when $\delta_k > 0$. The completed state is the sum of these two positive classical parts with weights $1 - \delta_k$ and δ_k . Let $\mathcal{Q}$ be the product of the two local classical channels sending each party's own abort symbol to zero and fixing its bit values. It is defined on every completed output and preserves each non-aborting ideal bit γ_k . Since the trace-norm distance between normalized states is at most two,

$$\|\mathcal{Q}(\tau_k) - \gamma_k\|_1 \leq (1 - \delta_k) \|\tau_k^{\text{ok}} - \gamma_k\|_1 + 2\delta_k.$$

The right side tends to zero under the stated hypotheses. This contradicts Theorem 5.3 for the completed bit-output protocol with the local replacement appended. The replacement uses only abort symbols actually available to the respective laboratories.

For the third assertion, denote the recovery channel for the k th embedding by $\mathcal{R}_k$. For any ideal bit γ_σ in the common space, applying this channel to Eve maps it to the ideal bit $\gamma_{\mathcal{R}_k(\sigma)}$ on the k th output space and recovers τ_k from $\hat{\tau}_k$. Contraction and the per-protocol gap therefore give

$$\|\hat{\tau}_k - \gamma_\sigma\|_1 \geq \|\tau_k - \gamma_{\mathcal{R}_k(\sigma)}\|_1 \geq \frac{1}{5}.$$

The triangle inequality now yields

$$\|\tau - \gamma_\sigma\|_1 \geq \frac{1}{5} - \|\hat{\tau}_k - \tau\|_1 \longrightarrow \frac{1}{5}.$$

Only the completed actual outputs need converge. The embeddings must retain and permit recovery of the full purification and public records, so they do not coarse-grain that information. The argument requires no convergence on the auxiliary probe inputs and no factorization statement for the limit itself. In particular, it does not assume that finite padded transcript laws converge in trace norm to a law of an infinite transcript. $\square$

6 Tensor geometry from singular directions

We now construct the PPT pair in Theorem 1.2. The geometric step uses a map on $M_4(\mathbb{C})$ with twenty specified singular outputs. It turns each singular output into a product vector in the kernel of a Choi matrix on $\mathbb{C}^{10} \otimes \mathbb{C}^{10}$. A condition on quadratic evaluations then excludes product vectors from the range.

For a vector x , write $P_x = xx^*$, without normalizing it. A projective direction $[x]$ is the complex line spanned by a nonzero vector x .

Proposition 6.1 (Tensor criterion). *Let $L : M_4(\mathbb{C}) \rightarrow M_4(\mathbb{C})$ be PPT. Suppose that $\mathcal{X} = \{x_1, \dots, x_{20}\} \subset \mathbb{C}^4 \setminus \{0\}$ represents twenty distinct projective directions and has the following properties:*

1. *rank $L(P_x) < 4$ for every $x \in \mathcal{X}$, and every nonzero homogeneous quadratic in $\mathbb{C}[x_0, x_1, x_2, x_3]$ vanishes on at most nine members of $\mathcal{X}$;*
2. *$L(P_y)$ is positive definite for some $y \in \mathbb{C}^4$.*

Then there are PPT maps $\Phi_1, \Phi_2 : M_{10}(\mathbb{C}) \rightarrow M_{10}(\mathbb{C})$, constructed explicitly from L , such that

$$Z = J(\Phi_2 \circ \Phi_1) \neq 0, \quad \text{ran } Z \cap \{u \otimes v : u, v \in \mathbb{C}^{10}\} = \{0\}.$$

In particular, $\Phi_2 \circ \Phi_1$ is not entanglement breaking.

Section 7 supplies an integer construction of L with these properties. We prove the criterion by using the symmetric square to record quadratic monomials and the exterior square to detect rank loss.

Proof. Symmetric and exterior tensors. Index an orthonormal basis of $\mathbb{C}^{10}$ by pairs $0 \leq i \leq j < 4$, and an orthonormal basis of $\mathbb{C}^6$ by pairs $0 \leq i < j < 4$, both in lexicographic order. Define isometries $U : \mathbb{C}^{10} \rightarrow \mathbb{C}^4 \otimes \mathbb{C}^4$ and $V : \mathbb{C}^6 \rightarrow \mathbb{C}^4 \otimes \mathbb{C}^4$ by

$$\begin{aligned} Ue_{ii} &= e_i \otimes e_i, & Ue_{ij} &= \frac{e_i \otimes e_j + e_j \otimes e_i}{\sqrt{2}} \quad (i < j), \\ Ve_{ij} &= \frac{e_i \otimes e_j - e_j \otimes e_i}{\sqrt{2}} \quad (i < j). \end{aligned} \tag{6.1}$$

Their ranges are the symmetric and antisymmetric tensor subspaces. Put

$$\hat{x} = U^*(x \otimes x) = \sum_i x_i^2 e_{ii} + \sqrt{2} \sum_{i < j} x_i x_j e_{ij} \quad (x \in \mathbb{C}^4). \tag{6.2}$$

Thus $U\hat{x} = x \otimes x$ and $UP_x U^* = P_x \otimes P_x$. The coordinates of $\hat{x}$ are the ten quadratic monomials with nonzero normalization factors. Hence the quadratic hypothesis is equivalent to the assertion that any ten vectors among $\{\hat{x} : x \in \mathcal{X}\}$ span $\mathbb{C}^{10}$: a failure to span would give a nonzero vector u with $u^* \hat{x} = 0$ at those ten points, and $u^* \hat{x}$ is a nonzero homogeneous quadratic.

Identify $\mathbb{C}^6$ with $\wedge^2 \mathbb{C}^4$ by $e_{ij} = e_i \wedge e_j$, with this basis orthonormal. Under this identification,

$$V(a \wedge b) = \frac{a \otimes b - b \otimes a}{\sqrt{2}}.$$

For $i < j$, let $k < l$ be the complementary indices, and let ϵ_{ijkl} be the sign of the ordering (i, j, k, l) of $(0, 1, 2, 3)$. Define the real signed permutation K on $\mathbb{C}^6$ by

$$K e_{ij} = \epsilon_{ijkl} e_{kl}. \quad (6.3)$$

Explicitly,

$$(K e_{01}, K e_{02}, K e_{03}, K e_{12}, K e_{13}, K e_{23}) = (e_{23}, -e_{13}, e_{12}, e_{03}, -e_{02}, e_{01}),$$

so $K = K^* = K^{-1}$. The conjugate operation $w \mapsto K \bar{w}$ is the complementary-pair exterior dual also used by Schliemann, Loss, and MacDonald for two fermions in a four-dimensional one-particle space [55, Section III, Equations (26)–(28)].

Define $S, R : M_{10}(\mathbb{C}) \rightarrow M_6(\mathbb{C})$ by

$$S(A) = V^*(L \otimes L)(U A U^*)V, \quad R(A) = K S(A)^\top K^*. \quad (6.4)$$

The embeddings and compressions are CP, so Lemma 2.1 makes S PPT. The map $T_6 \circ S$ is CP by this PPT property, and its output transpose is S , which is CP. Thus $T_6 \circ S$ is PPT, and the CP conjugation by K makes R PPT.

The composite Choi matrix. The rectangular maps give an endomorphism $R^\dagger \circ S \circ T_{10}$ of $M_{10}(\mathbb{C})$. To express it as the composition of two maps on that algebra, let $E : \mathbb{C}^6 \rightarrow \mathbb{C}^{10}$ be the real coordinate inclusion into the first six positions and define

$$\Phi_1(A) = E S(A^\top) E^*, \quad \Phi_2(B) = R^\dagger(E^* B E). \quad (6.5)$$

For Φ_1 , Lemma 2.1 makes $S \circ T_{10}$ CP by input/output transpose equivalence, and makes its output transpose $T_6 \circ S \circ T_{10}$ CP by simultaneous transposition of S . Thus $S \circ T_{10}$ is PPT, and the CP embedding by E preserves that property. The adjoint and CP-composition clauses of the same lemma make Φ_2 PPT. The formulas determine both maps explicitly once L is given. For example, if $L(A) = \sum_r Q_r A Q_r^*$, the adjoints in the construction are

$$\begin{aligned} L^\dagger(Y) &= \sum_r Q_r^* Y Q_r, \\ S^\dagger(Y) &= U^*(L^\dagger \otimes L^\dagger)(V Y V^*)U, \\ R^\dagger(Y) &= S^\dagger((K^* Y K)^\top). \end{aligned}$$

Since $E^* E = I_6$, $\Phi_2 \circ \Phi_1 = R^\dagger \circ S \circ T_{10}$. Its Choi matrix Z is positive. The convention (2.1) gives, for arbitrary matrices a, b of the appropriate sizes,

$$\text{tr}[(a \otimes b)J(F)] = \text{tr}[bF(a^\top)].$$

For Hermitian $a, b \in M_{10}(\mathbb{C})$, the Hilbert–Schmidt adjoint identity therefore gives

$$\text{tr}[(a \otimes b)Z] = \text{tr}[bR^\dagger(S(a))] = \text{tr}[R(b)S(a)]. \quad (6.6)$$

The Choi transpose has canceled the input transpose in (6.5). We will prove that the right side vanishes at $a = b = P_{\hat{x}}$ for each singular direction. Positivity of Z will then give the kernel vector $\hat{x} \otimes \hat{x}$. The positive definite output will separately show that Z is nonzero.

Singular outputs and kernel products. For any operator B on $\mathbb{C}^4$,

$$V^*(B \otimes B)V = \bigwedge^2 B.$$

Both sides send $a \wedge b$ to $Ba \wedge Bb$. If $B_x = L(P_x) \succeq 0$ and $W_x = \text{ran } B_x$, it follows that

$$S(P_{\hat{x}}) = \bigwedge^2 B_x, \quad \text{ran } S(P_{\hat{x}}) = \bigwedge^2 W_x. \quad (6.7)$$

Indeed, in an orthonormal eigenbasis of B_x , the exterior eigenvalues are $\lambda_i \lambda_j$ for $i < j$. This proves both positivity and the range identity. A Hermitian matrix D satisfies $D^\top = \overline{D}$, so transposition conjugates its range. Consequently

$$\text{ran } R(P_{\hat{x}}) = \overline{K \bigwedge^2 W_x}. \quad (6.8)$$

For exterior two-vectors w, u , let $[w \wedge u]_{0123}$ denote the coefficient of $e_0 \wedge e_1 \wedge e_2 \wedge e_3$. With the Hermitian inner product conjugate-linear in its first argument, the definition of K gives

$$\langle K\overline{w}, u \rangle = \sum_{i < j} w_{ij} \epsilon_{ijkl} u_{kl} = [w \wedge u]_{0123}. \quad (6.9)$$

Here $k < l$ is complementary to $i < j$ in each term. If $w, u \in \bigwedge^2 W$ and $\dim W < 4$, then $w \wedge u \in \bigwedge^4 W = 0$. Equations (6.7)–(6.9) show that $S(P_{\hat{x}})$ and $R(P_{\hat{x}})$ have orthogonal ranges for every $x \in \mathcal{X}$. This includes $\dim W_x < 2$, when both exterior ranges are zero. The conjugation in (6.9) is essential for this conclusion over complex subspaces.

For $x \in \mathcal{X}$, orthogonality of the two positive output ranges gives

$$(\hat{x} \otimes \hat{x})^* Z (\hat{x} \otimes \hat{x}) = \text{tr}[R(P_{\hat{x}})S(P_{\hat{x}})] = 0.$$

For a positive matrix, $v^* Z v = 0$ implies $Zv = 0$. Hence

$$\hat{x} \otimes \hat{x} \in \ker Z \quad (x \in \mathcal{X}). \quad (6.10)$$

For the input y in the second hypothesis, $S(P_{\hat{y}}) = \bigwedge^2(L(P_y))$ is positive definite, and so is $R(P_{\hat{y}}) = K S(P_{\hat{y}})^\top K^*$. The trace of their product is strictly positive. Equation (6.6) at $a = b = P_{\hat{y}}$ therefore proves $Z \neq 0$.

Excluding product vectors from the range. It remains to exclude a product vector from the range. The following zero-set argument uses the product-exclusion principle of Bennett and coauthors [10, Lemma 1], in the nonorthogonal form given by Pittenger [49, Section IV, Lemma 3], and is related to Parthasarathy's polynomial interpolation construction [45, Proposition 1.3]. Suppose that a nonzero product $u \otimes v$ belongs to $\text{ran } Z$. Since Z is Hermitian, its range is the orthogonal complement of its kernel. Equation (6.10) implies

$$p_u(x)p_v(x) = 0 \quad (x \in \mathcal{X}), \quad p_u(x) = u^* \hat{x}, \quad p_v(x) = v^* \hat{x}.$$

For example,

$$p_u(x) = \sum_i \overline{u_{ii}} x_i^2 + \sqrt{2} \sum_{i < j} \overline{u_{ij}} x_i x_j.$$

This is a homogeneous quadratic in x ; only its fixed coefficients are conjugated. Its distinct monomials make it nonzero when $u \neq 0$. The same holds for p_v , since $v \neq 0$. Each polynomial therefore vanishes on at most nine members of $\mathcal{X}$. Their zero sets cannot cover all twenty members, because $9 + 9 < 20$. This contradiction proves that $\text{ran } Z$ contains no nonzero product vector.

By Lemma 2.3, the nonzero positive matrix Z is nonseparable. Lemma 2.2 then shows that the composite is not entanglement breaking. $\square$

7 An integer pencil and its exact certificate

This section verifies every hypothesis of Proposition 6.1 for an explicit map $L : M_4(\mathbb{C}) \rightarrow M_4(\mathbb{C})$. Four integer 6-by-4 matrices $M_0, \dots, M_3$ define L and the pencil $M(x) = \sum_{i=0}^3 x_i M_i$. The maximal minors of this pencil identify all singular projective directions. Exact reductions at three primes, together with the quotient and Galois arguments below, control quadratic evaluations on every ten of those directions.

7.1 The explicit matrix pencil

Use indices 0, 1, 2, 3 and define real 6-by-4 matrices

$$\begin{aligned} M_0 &= \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, & M_1 &= \begin{pmatrix} -12 & 0 & 0 & 0 \\ 0 & -6 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 12 \\ -6 & 0 & 6 & 6 \\ -6 & -6 & 6 & 0 \end{pmatrix}, \\ M_2 &= \begin{pmatrix} 0 & 6 & -2 & 0 \\ 6 & 6 & 0 & 2 \\ -2 & 0 & 10 & 0 \\ 0 & 2 & 0 & 24 \\ 0 & 0 & 0 & 6 \\ 0 & -6 & 6 & -6 \end{pmatrix}, & M_3 &= \begin{pmatrix} 0 & 0 & -4 & -3 \\ 0 & -2 & -3 & -4 \\ -4 & -3 & 11 & 6 \\ -3 & -4 & 6 & 25 \\ 6 & 6 & 0 & 0 \\ 0 & -6 & 6 & 6 \end{pmatrix}. \end{aligned} \tag{7.1}$$

For $x = (x_0, x_1, x_2, x_3) \in \mathbb{C}^4$, put $M(x) = \sum_{i=0}^3 x_i M_i$ and define the complex-linear map L on matrix units by

$$L(E_{ij}) = M_i^\top M_j \quad (0 \leq i, j < 4). \tag{7.2}$$

Lemma 7.1. *The map $L : M_4(\mathbb{C}) \rightarrow M_4(\mathbb{C})$ in (7.2) is PPT. For every $x \in \mathbb{C}^4$,*

$$L(P_x) = M(x)^\top \overline{M(x)} = M(x)^* M(x), \quad \text{rank } L(P_x) = \text{rank } M(x). \tag{7.3}$$

Moreover, $L(P_{e_0}) = 36I_4$.

Proof. Direct integer multiplication gives

$$M_i^\top M_j = M_j^\top M_i \quad (0 \leq i, j < 4). \tag{7.4}$$

The exact checker in Appendix A verifies these equalities. For each row r , let Q_r be the 4-by-4 matrix whose i th column is the transpose of row r of M_i . Comparing entries gives $L(A) = \sum_{r=1}^6 Q_r A Q_r^*$, so L is CP. Equation (7.4) also makes every $M_i^\top M_j$ symmetric. Therefore $T_4 \circ L = L$, proving the PPT property.

For complex x , expansion of (7.2) gives

$$L(P_x) = \sum_{i,j} x_i \overline{x_j} M_i^\top M_j = M(x)^\top \overline{M(x)}.$$

Interchanging i, j and using (7.4) gives the second expression in (7.3). Since $v^* M(x)^* M(x) v = \|M(x)v\|^2$, the kernel of this Gram matrix is $\ker M(x)$, which proves the rank equality. Finally, $M_0^\top M_0 = 36I_4$. $\square$

The positive-definite input required by the tensor criterion is now available. The remaining assertion concerns every complex rank-loss direction of the pencil.

Proposition 7.2 (Twenty rank-loss directions with independent quadratic evaluations). *The projective directions $[x] \in \mathbb{P}^3(\mathbb{C})$ for which $\text{rank } M(x) < 4$ consist of exactly twenty distinct points. Each has a unique representative $x = (1, t_1, t_2, t_3)$. Every nonzero homogeneous quadratic in $\mathbb{C}[x_0, x_1, x_2, x_3]$ vanishes on at most nine of these representatives.*

The equation $M(x)y = 0$ with $x, y \neq 0$ is related to the bilinear kernel geometry studied by Hansen, Hauge, Myrheim, and Sollid for 4-by-4 PPT states whose matrix and partial transpose both have rank six [27, Sections 2.2 and 7]. Their generic count concerns twenty product rays $x \otimes y$. The quadratic condition here concerns the vectors $x \otimes x$ for this specific pencil, and the following argument proves it from exact data.

Proof of Proposition 7.2. The maximal-minor relations first show that the affine quotient is nonzero and spanned by twenty monomials. We then construct a formal multiplication matrix. Its irreducible characteristic polynomial, certified modulo 41, proves that the spanning monomials are independent and that the quotient is a degree-twenty field. This identifies all twenty complex directions. Finally, reductions at 131 and 139 give the full symmetric Galois action, which transports one independent set of ten quadratic evaluations to every such set. The three general algebraic facts used along the way are stated at their uses; their proofs are collected in Appendix A.2.

The checker in Appendix A supplies the integer block identities and the finite modular data used in these stages. It does not enumerate the twenty complex directions or their ten-point determinants. The quotient, projective, and Galois deductions are proved below.

7.2 The maximal minors and a nonzero quotient

Because $M(x)$ has four columns, the condition $\text{rank } M(x) < 4$ is equivalent to simultaneous vanishing of its fifteen maximal minors. For each four-element subset $\rho \subset \{1, \dots, 6\}$, taken in lexicographic order with its rows in increasing order, let $\Delta_\rho(x)$ be the determinant of those rows of $M(x)$. Put $p_\rho(t) = \Delta_\rho(1, t_1, t_2, t_3)$ and define

$$I = (p_\rho)_\rho \subset \mathbb{Q}[t_1, t_2, t_3], \quad \mathcal{A} = \mathbb{Q}[t_1, t_2, t_3]/I.$$

These equations describe the singular directions in the chart $x_0 = 1$. We have not yet proved that I is proper, or that this chart contains all the singular directions.

Order monomials first by increasing total degree d and, within degree d , by the exponent triples

$$(a, b, d - a - b), \quad a = 0, \dots, d, \quad b = 0, \dots, d - a, \quad (7.5)$$

in the displayed nested order. Let v be the column of the twenty monomials of degree at most three, and let w be the column of the fifteen monomials of degree four. The coefficients of the minors define integer matrices D, H by

$$(p_\rho(t))_\rho = Dv(t) + Hw(t), \quad D \in \mathbb{Z}^{15 \times 20}, \quad H \in \mathbb{Z}^{15 \times 15}. \quad (7.6)$$

The exact coefficient calculation in Appendix A gives $\det H \equiv 40 \pmod{41}$, so H is invertible over $\mathbb{Q}$. Define $C = -H^{-1}D$. In the quotient $\mathcal{A}$, Equation (7.6) gives the relations $w = Cv$, expressing each degree-four monomial as a linear combination of the twenty lower monomials.

The relations $w = Cv$ must describe a nonzero quotient before they can be used to count its points. We first verify this and show that the low monomials span. The existence argument uses the following intersection fact for $\mathbb{P}^3 \times \mathbb{P}^3$.

Lemma 7.3 (Six bilinear equations). *Any six complex bilinear forms on $\mathbb{C}^4 \times \mathbb{C}^4$ have a common zero (x, y) with $x \neq 0$ and $y \neq 0$.*

The proof is given in Appendix A.2.

Apply the lemma to the six rows of $M(x)y$. It gives $x, y \neq 0$ with $M(x)y = 0$, so all $\Delta_\rho(x)$ vanish.

The homogeneous degree-four part of $p_\rho(t)$ is $\Delta_\rho(0, t_1, t_2, t_3)$. Equation (7.6) therefore gives

$$(\Delta_\rho(0, t))_\rho = Hw(t).$$

If a common zero had $x_0 = 0$, invertibility of H would imply $w(t) = 0$. The entries t_1^4, t_2^4, t_3^4 occur in w , so $t = 0$, contradicting $x \neq 0$. Thus every projective common zero lies in the chart $x_0 = 1$. Rescaling the zero furnished by the lemma gives an affine common zero of the p_ρ , proving that I is proper and $\mathcal{A} \neq 0$. The same argument has excluded all points at infinity.

In $\mathcal{A}$, the relations $w = Cv$ show that the twenty low monomials span. In a monomial of total degree at least four, substitute for any degree-four factor. Each resulting term has strictly smaller total degree. Induction reduces every monomial to a linear combination of the low monomials; this proves spanning without assuming unique reductions.

7.3 A formal multiplication operator

Multiplication matrices are a standard way to encode finite polynomial systems; see Sottile [57, Section 2.3]. The quotient $\mathcal{A}$ is now known to be nonzero and spanned by the twenty low monomials, but their independence remains to be proved. Take a formal twenty-dimensional rational vector space with basis e_α indexed by exponent triples $|\alpha| \leq 3$, in the order (7.5). Define an endomorphism N of this formal space and its characteristic polynomial by

$$Ne_\alpha = \begin{cases} e_{\alpha+(1,0,0)}, & |\alpha| \leq 2, \\ \sum_{|\beta| \leq 3} C_{\alpha+(1,0,0),\beta} e_\beta, & |\alpha| = 3, \end{cases} \quad f(z) = \det(zI_{20} - N). \quad (7.7)$$

The row index of C in the second line is the indicated degree-four monomial. Let $\pi : \mathbb{Q}^{20} \rightarrow \mathcal{A}$ send e_α to the class of t^α , and let μ_1 be multiplication by the class of t_1 . The spanning result makes π surjective. By the definition of N ,

$$\pi N = \mu_1 \pi. \quad (7.8)$$

In particular, $\ker \pi$ is an N -invariant rational subspace. We will use irreducibility of f to show that this kernel is zero, which will identify N with multiplication by t_1 .

For a prime p with $\det H \not\equiv 0 \pmod{p}$, set

$$\mathbb{Z}_{(p)} = \{a/b : a, b \in \mathbb{Z}, p \nmid b\}.$$

The identity $H^{-1} = \text{adj}(H)/\det H$ shows that the entries of C, N and the coefficients of the monic polynomial f lie in $\mathbb{Z}_{(p)}$. In particular their denominators are units at p , and reduction gives $\bar{f} = \det(zI_{20} - \bar{N}) \in \mathbb{F}_p[z]$. This is a statement of integrality at the specified prime; no global integrality of the coefficients of f is needed.

For $k \geq 1$, put

$$d_p(k) = \deg \gcd(\bar{f}, z^{p^k} - z).$$

The exact calculation in Appendix A gives the computed columns of Table 1. Its output also records all twenty-one characteristic coefficients and all twenty gcd degrees at each prime.

Table 1: Computed modular data and the factor degrees deduced from them.

p	$\det H \bmod p$	computed gcd degrees	deduced factor degrees
41	40	$d_{41}(k) = 0$ ($1 \leq k < 20$), $d_{41}(20) = 20$	20
131	23	$d_{131}(1) = 1$, $d_{131}(2) = 3$, $d_{131}(17) = 18$	1, 2, 17
139	9	$d_{139}(1) = 1$, $d_{139}(19) = 20$	1, 19

The determinant residues justify all three local reductions. The primes exceed twenty, so the divisions in the Newton identities used to compute the characteristic polynomials are valid. Their primality follows by trial division by the primes at most eleven.

We explain the last column of the table. Over $\mathbb{F}_p$ the polynomial $z^{p^k} - z$ is squarefree, since its derivative is -1 . Its irreducible factors are precisely the monic irreducible polynomials whose degrees divide k : a root of an irreducible polynomial of degree s has a Frobenius orbit of length s . Hence $d_p(k)$ is the sum of the degrees of the *distinct* irreducible factors of $\bar{f}$ whose degrees divide k . At $p = 41$, a factor of degree $s < 20$ would make $d_{41}(s)$ positive, so $\bar{f}$ is irreducible of degree twenty. At $p = 131$, the three displayed values force one factor of degree one, one of degree two, and one of degree seventeen. These degrees already sum to twenty, so there are no further or repeated factors. The two values at $p = 139$ similarly force one factor of degree one and one of degree nineteen, without repetition. All three reductions are squarefree.

The following elementary reduction lemma lets us infer rational irreducibility without assuming that f has integer coefficients. Its p -adic form is used in the proof of Lemma 7.5 in Appendix A.2.

Lemma 7.4 (Monic local reduction). *Let p be a prime, and let the coefficient field and its valuation ring be one of the pairs*

$$(\mathbb{L}, R_p) = (\mathbb{Q}, \mathbb{Z}_{(p)}) \quad \text{or} \quad (\mathbb{L}, R_p) = (\mathbb{Q}_p, \mathbb{Z}_p).$$

If a monic polynomial $P \in R_p[z]$ factors as $P = gh$ with monic $g, h \in \mathbb{L}[z]$, then $g, h \in R_p[z]$, and their reductions modulo p retain their degrees, including degree zero. Consequently, if P has positive degree and irreducible reduction in $\mathbb{F}_p[z]$, then P is irreducible over $\mathbb{L}$.

The proof is given in Appendix A.2.

Applying the lemma to $f \in \mathbb{Z}_{(41)}[z]$ and its irreducible reduction proves that f is irreducible over $\mathbb{Q}$.

7.4 The coordinate field and its twenty points

We can now prove independence of the twenty spanning monomials. An endomorphism with irreducible characteristic polynomial has no nonzero proper invariant subspace: a basis adapted to such a subspace makes the matrix block triangular and factors its characteristic polynomial into those of the restriction and quotient. Since π is surjective and $\mathcal{A} \neq 0$, its kernel is proper. Irreducibility of f forces $\ker \pi = 0$. The twenty low monomials are therefore a basis of $\mathcal{A}$, and N is its multiplication matrix for t_1 .

By Cayley–Hamilton the minimal polynomial of N divides f . It is nonconstant, so irreducibility of f makes it equal to f . In a nonzero unital algebra, a polynomial in an element vanishes exactly when the same polynomial in its multiplication operator vanishes; the reverse implication follows by applying the operator to 1. Hence the subalgebra generated by t_1 is $\mathbb{Q}[z]/(f)$ and has dimension twenty. It is already all of $\mathcal{A}$:

$$\mathcal{A} \simeq \mathbb{Q}[z]/(f), \quad z \longmapsto t_1. \quad (7.9)$$

In particular $\mathcal{A}$ is a field. Write the classes of t_2, t_3 as $h_2(t_1), h_3(t_1)$ with $h_2, h_3 \in \mathbb{Q}[z]$ of degree below twenty.

The irreducible polynomial f is separable in characteristic zero. Let its distinct complex roots be $\lambda_1, \dots, \lambda_{20}$. Extending scalars in (7.9) and applying the Chinese remainder theorem gives

$$\mathbb{C}[t_1, t_2, t_3]/I\mathbb{C}[t_1, t_2, t_3] \simeq \mathbb{C}[z]/(f) \simeq \prod_{j=1}^{20} \mathbb{C}. \quad (7.10)$$

The factors are evaluations at the distinct triples

$$t^{(j)} = (\lambda_j, h_2(\lambda_j), h_3(\lambda_j)). \quad (7.11)$$

Conversely, any complex common zero of the p_ρ defines a unital $\mathbb{C}$ -algebra homomorphism from the left side of (7.10) to $\mathbb{C}$, and such a homomorphism selects one factor of the product. Thus the triples in (7.11) are all the affine common zeros. The product of fields also shows that the complex ideal is radical, so these twenty points are reduced. Together with the exclusion of $x_0 = 0$, this identifies all complex projective singular directions of the pencil.

7.5 Local Frobenius and quadratic evaluations

It remains to prove that no nonzero quadratic vanishes on ten of the directions. We will show that the Galois group of the splitting field permutes the twenty roots arbitrarily. This transports independence for one set of ten evaluations to every set of ten.

The following is the locally integral form of the Frobenius factorization principle, often called Dedekind's theorem. Milne states the monic integer case [43, Theorem 4.28]. We give the local rational argument needed for the coefficients of f .

Lemma 7.5 (Locally integral Frobenius). *Let $F \in \mathbb{Q}[z]$ be monic, separable, and of positive degree, with coefficients in $\mathbb{Z}_{(p)}$. If its reduction in $\mathbb{F}_p[z]$ is squarefree with irreducible-factor degrees $s_1, \dots, s_h$, then the Galois group of its rational splitting field contains an element with cycle lengths $s_1, \dots, s_h$ on the roots.*

The proof is given in Appendix A.2.

Let $\mathbb{K} \subset \mathbb{C}$ be a splitting field of f , and let $G = \text{Gal}(\mathbb{K}/\mathbb{Q})$ act on its twenty roots. Applying Lemma 7.5 to the reductions at 131 and 139 gives elements of cycle types $(1, 2, 17)$ and $(1, 19)$. Irreducibility of f makes this action transitive. A nineteen-cycle fixes one root and is transitive on the other nineteen, so the stabilizer of that root is transitive on its complement. Transitivity conjugates this property to every point stabilizer; hence G is transitive on ordered pairs of distinct roots, or two-transitive. The seventeenth power of an element of type $(1, 2, 17)$ is a transposition. Two-transitivity sends its two entries to any ordered pair, and conjugation therefore gives every transposition. Thus

$$G = S_{20}. \quad (7.12)$$

This is the elementary permutation criterion recorded in Milne [43, Lemma 4.31]. The group in (7.12) is the splitting-field group acting on the twenty roots.

Form the 20-by-10 evaluation matrix over $\mathbb{K}$

$$\mathcal{E}_{j,\alpha} = (t^{(j)})^\alpha, \quad 1 \leq j \leq 20, \quad |\alpha| \leq 2.$$

Its entries lie in $\mathbb{K}$ by (7.11). The ten column monomials belong to the basis of $\mathcal{A}$ already proved above. They remain independent after scalar extension to $\mathbb{K}$, where the split algebra is $\mathbb{K}^{20}$ through

evaluation at the twenty triples. Hence $\mathcal{E}$ has column rank ten over $\mathbb{K}$, so some ten-row square submatrix has nonzero determinant.

Every $\sigma \in G$ permutes the triples in (7.11) according to its permutation of the roots, since h_2, h_3 have rational coefficients. In particular $\sigma(\mathcal{E}_{j,\alpha}) = \mathcal{E}_{\sigma(j),\alpha}$. The group S_{20} is transitive on ten-element subsets. Applying a suitable σ to the known nonzero determinant therefore gives, up to a row-order sign, the determinant for any prescribed ten rows. A field automorphism preserves nonzero elements, so every such determinant is nonzero in $\mathbb{K}$ and hence in $\mathbb{C}$. The Galois action has been used only on algebraic determinants; their nonvanishing gives invertibility for arbitrary complex coefficient vectors.

Every homogeneous quadratic has a unique expression

$$q(x_0, x_1, x_2, x_3) = \sum_{|\alpha| \leq 2} a_\alpha x_0^{2-|\alpha|} x_1^{\alpha_1} x_2^{\alpha_2} x_3^{\alpha_3}.$$

Its dehomogenization $q(1, t)$ is nonzero exactly when q is nonzero. If it vanished at ten triples, the corresponding invertible evaluation matrix would force every $a_\alpha = 0$. Thus it vanishes on at most nine triples. The identification of all common zeros above, the exclusion of $x_0 = 0$, and the equivalence between vanishing of the maximal minors and $\text{rank } M(x) < 4$ now prove Proposition 7.2. $\square$

Proof of Theorem 1.2. Take L from (7.2). Lemma 7.1 supplies its PPT property, the rank identity, and the positive-definite input $L(P_{e_0}) = 36I_4$. Proposition 7.2 supplies the twenty distinct directions and the quadratic bound. Proposition 6.1 therefore applies, with the explicit maps in (6.5) determined by the integer matrices (7.1).

The positive test also gives an exact nonzero Choi value. Since $\hat{e}_0 = e_{00}$, Equations (6.4) and (6.6) yield

$$S(P_{e_{00}}) = R(P_{e_{00}}) = 36^2 I_6, \quad (e_{00} \otimes e_{00})^* Z(e_{00} \otimes e_{00}) = 6 \cdot 36^4 = 10,077,696 > 0.$$

Thus the constructed pair has the asserted nonzero product-free composite Choi range, and its composite is not entanglement breaking. $\square$

8 The state and a trace-preserving channel

We now combine the two independent inputs to the main state theorem: the operational obstruction for the represented class and the product-free Choi range of the explicit dimension-ten pair.

Proof of Theorem 1.1. Take Φ_1, Φ_2 from (6.5) and put $Z = J(\Phi_2 \circ \Phi_1)$. Theorem 1.2 proves that Z is nonzero and that its range contains no nonzero product vector. It is positive because $\Phi_2 \circ \Phi_1$ is CP. Thus $\lambda = \text{tr } Z > 0$, and $\rho = Z/\lambda$ is a density operator on $\mathbb{C}^{10} \otimes \mathbb{C}^{10}$ with the same range as Z . Lemma 2.3 makes ρ entangled.

Both maps are PPT. Lemma 2.6, with $a = b = c = 10$, gives

$$\rho = (\Phi_1^\# \otimes \Phi_2) \left(\frac{\Omega_{10} \Omega_{10}^*}{\lambda} \right) \in \mathcal{C}.$$

In particular, the representing vector is $\Omega_{10}/\sqrt{\lambda}$; normalization is by the actual Choi trace. Theorem 5.3 now gives (1.1) for every $n \geq 1$ and every protocol and ideal bit in its stated scope, including the purification and complete public record. The same theorem gives $K_D(\rho) = 0$. All matrices defining ρ are specified by (7.1) and (6.5), so the state is explicit. $\square$

The maps Φ_1, Φ_2 need not preserve trace. The next proposition converts any square PPT pair into a trace-preserving PPT channel, while retaining the pair's composite Choi matrix in a local corner. The two-block switch follows Christandl, Müller-Hermes, and Wolf [16, Section IV.A]. The enlarged-input absorbing flag follows Filippov's trace-preserving extension [21, Proposition 3, Equation (15)].

Proposition 8.1. *Let $\Psi_1, \Psi_2 : M_m(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ be PPT maps. There are explicitly defined constants $c_1, c_2 > 0$, a trace-preserving PPT channel $\Theta : M_{2m+1}(\mathbb{C}) \rightarrow M_{2m+1}(\mathbb{C})$, and a real coordinate isometry $W_0 : \mathbb{C}^m \rightarrow \mathbb{C}^{2m+1}$ such that*

$$(W_0^* \otimes W_0^*)J(\Theta \circ \Theta)(W_0 \otimes W_0) = c_1 c_2 J(\Psi_2 \circ \Psi_1). \quad (8.1)$$

If $J(\Psi_2 \circ \Psi_1)$ is nonseparable, then $\Theta \circ \Theta$ is not entanglement breaking.

Proof. Define

$$A_i = \Psi_i^\dagger(I_m), \quad c_i = \frac{1}{1 + \text{tr } A_i}, \quad F_i = I_m - c_i A_i \quad (i = 1, 2). \quad (8.2)$$

The adjoint of a CP map is CP, so $A_i \succeq 0$. Since $A_i \preceq (\text{tr } A_i)I_m$, we have $F_i \succeq (1 - c_i \text{tr } A_i)I_m = c_i I_m \succ 0$. The adjoint identity, using that A_i is Hermitian, gives

$$\text{tr } \Psi_i(B) = \text{tr}(A_i B), \quad \text{tr } B - c_i \text{tr } \Psi_i(B) = \text{tr}(F_i B)$$

for every complex matrix B . Hence $c_i \Psi_i$ is trace nonincreasing on positive inputs, and the positive effect F_i records the missing trace.

Use the standard-coordinate decomposition

$$\mathbb{C}^{2m+1} = \mathbb{C}_0^m \oplus \mathbb{C}_1^m \oplus \mathbb{C}e_\star,$$

and let W_0, W_1 be the real coordinate inclusions. For $X \in M_{2m+1}(\mathbb{C})$ write $X_{jj} = W_j^* X W_j$, $x_\star = e_\star^* X e_\star$, and $P_\star = e_\star e_\star^*$. Set

$$\begin{aligned} \Theta(X) &= c_2 W_0 \Psi_2(X_{11}) W_0^* + c_1 W_1 \Psi_1(X_{00}) W_1^* \\ &\quad + (\text{tr}(F_1 X_{00}) + \text{tr}(F_2 X_{11}) + x_\star) P_\star. \end{aligned} \quad (8.3)$$

The formula discards the off-diagonal input blocks. The two moving branches are CP compositions. For the flag branches, take spectral decompositions $F_i = \sum_r \lambda_{ir} v_{ir} v_{ir}^*$ and set

$$K_{ir} = \sqrt{\lambda_{ir}} e_\star v_{ir}^* W_{i-1}^* \quad (i = 1, 2).$$

Then $\text{tr}(F_i X_{i-1, i-1}) P_\star = \sum_r K_{ir} X K_{ir}^*$. The remaining flag term is $P_\star X P_\star$. Thus Θ is CP. Since $c_i A_i + F_i = I_m$, for every complex X ,

$$\begin{aligned} \text{tr } \Theta(X) &= \text{tr}[(c_1 A_1 + F_1) X_{00}] + \text{tr}[(c_2 A_2 + F_2) X_{11}] + x_\star \\ &= \text{tr } X_{00} + \text{tr } X_{11} + x_\star = \text{tr } X. \end{aligned}$$

The discarded blocks have zero trace, so Θ is trace preserving.

Because $W_0, W_1, e_\star$ have real coordinates, output transposition in (8.3) replaces Ψ_i by $T_m \circ \Psi_i$ in the moving branches and fixes the flag output. Coordinate transpose is complex-linear: it does not conjugate the possibly complex scalar multiplying $P_\star$. The effects F_i need not be real. Each

transposed moving branch is CP by the PPT hypothesis, and the flag branches are CP as shown above. Therefore $T_{2m+1} \circ \Theta$ is CP, proving that Θ is PPT.

The flag is absorbing: $\Theta(P_\star) = P_\star$. Applying (8.3) twice to an input on the first block gives, for every $B \in M_m(\mathbb{C})$,

$$\begin{aligned} \Theta^2(W_0 B W_0^*) &= c_1 c_2 W_0 \Psi_2(\Psi_1(B)) W_0^* \\ &\quad + (\text{tr}(F_1 B) + c_1 \text{tr}(F_2 \Psi_1(B))) P_\star. \end{aligned}$$

Compression to that block consequently gives

$$W_0^* \Theta^2(W_0 B W_0^*) W_0 = c_1 c_2 \Psi_2(\Psi_1(B)). \quad (8.4)$$

Expanding in the coordinate matrix units of the input turns (8.4) into (8.1). The coordinate qualification fixes the reference factor: for general complex isometries W, V and a compatible map Γ ,

$$J(B \mapsto V^* \Gamma(W B W^*) V) = (W^T \otimes V^*) J(\Gamma) (\overline{W} \otimes V).$$

Here W_0 is real, so the reference factor is exactly the one displayed in (8.1).

Local compression preserves separability, since it takes a positive product $A \otimes B$ to $(W_0^* A W_0) \otimes (W_0^* B W_0)$. Thus separability of $J(\Theta^2)$ would imply separability of the positive multiple $c_1 c_2 J(\Psi_2 \circ \Psi_1)$ in (8.1). This proves the last assertion. $\square$

Proof of Theorem 1.3. Apply Proposition 8.1 with $m = 10$ and $\Psi_i = \Phi_i$ from (6.5). Their composite Choi matrix is nonseparable by Theorem 1.2 and Lemma 2.3. Formula (8.3) therefore gives the required PPT channel on $M_{21}(\mathbb{C})$. $\square$

The product-free range assertion is used for the compressed matrix Z . The full channel Choi matrix has a product contribution from the absorbing flag: since Θ annihilates off-diagonal blocks involving $e_\star$ and $\Theta(P_\star) = P_\star$, the matrix $P_\star \otimes P_\star$ is a direct summand of $J(\Theta^2)$. The channel conclusion is the nonseparability of the full Choi matrix, certified by its entangled corner.

9 An alternative construction from projection overlaps

The dimension-ten construction is complete. Sections 9 and 10 now produce another entangled member of $\mathcal{C}$ in much larger finite dimensions. The real construction has an additional symmetry: the same trace-preserving rectangular PPT channel acts on both halves of a maximally entangled state. These are parallel local actions on two subsystems, distinct from the self-composition of a square channel in Theorem 1.3.

The mechanism also differs from the dimension-ten range obstruction. The input consists of real unit vectors and projective measurements. The vectors produce a large sum of correlations, whereas zero trace overlaps among the projections restrict the supports of the product vectors in a hypothetical separable decomposition. These two bounds will conflict. Trace Gram matrices of positive factors also occur in completely positive semidefinite factorizations [51]; here the zero pattern of such trace pairings supplies the support restriction.

For self-adjoint projections E, F , their trace overlap is always a nonnegative real number:

$$\text{tr}(EF) = \text{tr}(FEF) = \text{tr}((EF)^* EF) \geq 0. \quad (9.1)$$

In particular, it vanishes exactly when $EF = 0$. This fact does not require E and F to commute.

Proposition 9.1 (Rectangular projection criterion). *Let Q, K, D be positive integers, let $\mathcal{Q}$ be a set of Q elements, and let $u_q \in \mathbb{R}^K$, $q \in \mathcal{Q}$, be unit vectors. Put $c_{qr} = u_q^\top u_r$. For each q , let $\{P_i : i \in \mathcal{I}_q\}$ be a finite projective measurement on $\mathbb{C}^D$: its members are self-adjoint projections, they are mutually orthogonal, and their sum is I_D . Zero projections are allowed. Take the label sets $\mathcal{I}_q$ disjoint, put $\mathcal{I} = \bigsqcup_{q \in \mathcal{Q}} \mathcal{I}_q$, and set $n = |\mathcal{I}|$. Suppose that*

1. $P_i P_j = P_j P_i$ whenever $i \in \mathcal{I}_q$, $j \in \mathcal{I}_r$, and $c_{qr} \neq 0$;
2. there is a real number $\kappa < Q/K$ such that every $J \subseteq \mathcal{I}$ satisfying $\text{tr}(P_i P_j) > 0$ for all distinct $i, j \in J$ has $|J| \leq \kappa$.

After ordering $\mathcal{I}$, define the complex-linear map $\Psi : M_n(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ on matrix units by

$$H_{ij} = c_{qr} P_i P_j, \quad \Psi(E_{ij}) = H_{ij} \quad (i \in \mathcal{I}_q, j \in \mathcal{I}_r). \quad (9.2)$$

Then Ψ and its Hilbert–Schmidt adjoint $\Psi^\dagger : M_D(\mathbb{C}) \rightarrow M_n(\mathbb{C})$ are PPT. Moreover,

$$C := J(\Psi^\dagger \circ \Psi) \quad \text{is nonseparable over } \mathbb{C}^n \otimes \mathbb{C}^n, \quad \text{tr } C = Q^2 D. \quad (9.3)$$

Proof. We first prove that Ψ is PPT. To establish nonseparability, let B be the compression of C to the span of the vectors $e_i \otimes e_i$, expressed in that basis, and let $\mathbf{1} = (1, \dots, 1)^\top$. We will compare two bounds on the sum of its entries. The vector Gram matrix gives

$$\mathbf{1}^* B \mathbf{1} \geq \frac{Q}{K} \text{tr } B,$$

whereas the zero tensor-basis diagonal entries would force a separable decomposition of C to satisfy $\mathbf{1}^* B \mathbf{1} \leq \kappa \text{tr } B$. We will also have $\text{tr } B = QD > 0$, so the strict inequality $\kappa < Q/K$ will make these bounds incompatible.

The block Gram matrix and its partial transposes. Regard each u_q also as a vector in $\mathbb{C}^K$. For arbitrary $w_i \in \mathbb{C}^D$, self-adjointness of the projections gives

$$\sum_{i,j \in \mathcal{I}} \langle w_i, H_{ij} w_j \rangle = \left\| \sum_{q \in \mathcal{Q}} \sum_{i \in \mathcal{I}_q} u_q \otimes P_i w_i \right\|^2 \geq 0. \quad (9.4)$$

Thus the block matrix $H = [H_{ij}]_{i,j \in \mathcal{I}} = J(\Psi)$ is positive semidefinite. Lemma 2.2 implies that Ψ is CP.

Since $c_{qr} = c_{rq}$ is real, the first hypothesis gives

$$H_{ij} = H_{ji} = H_{ij}^*. \quad (9.5)$$

Indeed, if $c_{qr} \neq 0$, the two projections commute; if $c_{qr} = 0$, both blocks vanish and no commutation is needed. Let $\Gamma_1 = T_n \otimes \text{id}_D$ and $\Gamma_2 = \text{id}_n \otimes T_D$ denote the partial transposes on the input and output Choi factors. Block symmetry gives

$$H^{\Gamma_1} = H, \quad H^{\Gamma_2} = H^\top \succeq 0. \quad (9.6)$$

For the second identity, full transpose is the product of the two partial transposes. Full transpose preserves complex positive semidefiniteness: from $H = R^* R$ we get $H^\top = (\bar{R})^* \bar{R}$. The Choi identities in Lemma 2.2 now show that $T_D \circ \Psi$ is CP. Hence Ψ is PPT.

Lemma 2.1 gives complete positivity and the PPT property for $\Psi^\dagger$. We will also use a more specific identity. Testing the defining adjoint relation on matrix units gives, for every $A \in M_D(\mathbb{C})$,

$$[\Psi^\dagger(A)]_{ij} = \text{tr}(H_{ij}^* A). \quad (9.7)$$

Equation (9.5) therefore yields

$$T_n \circ \Psi^\dagger = \Psi^\dagger \quad (9.8)$$

on all complex matrices, including non-Hermitian ones.

The tensor-basis diagonal and a compression. The composition $\Psi^\dagger \circ \Psi$ is CP, so $C \succeq 0$. Index its rows and columns by $\mathcal{I} \times \mathcal{I}$. Equation (9.7) gives the fully indexed formula

$$C_{(i,a),(j,b)} = \text{tr}(H_{ab}^* H_{ij}). \quad (9.9)$$

Since $c_{qq} = 1$ and $P_i^2 = P_i$, we have $H_{ii} = P_i$. The diagonal entries with indices (i, j) , and the entries of the compression B , are consequently

$$C_{(i,j),(i,j)} = \text{tr}(P_i P_j), \quad (9.10)$$

$$B_{ij} := C_{(i,i),(j,j)} = \text{tr}(H_{ij}^* H_{ij}) = c_{qr}^2 \text{tr}(P_i P_j) \quad (i \in \mathcal{I}_q, j \in \mathcal{I}_r). \quad (9.11)$$

The last equality follows from $\text{tr}(P_j P_i P_i P_j) = \text{tr}(P_i P_j)$, by idempotence and cyclicity of trace. It remains valid without commutation and when $c_{qr} = 0$.

For each pair q, r , completeness of the two measurements gives

$$\sum_{i \in \mathcal{I}_q} \sum_{j \in \mathcal{I}_r} \text{tr}(P_i P_j) = \text{tr} \left[\left(\sum_{i \in \mathcal{I}_q} P_i \right) \left(\sum_{j \in \mathcal{I}_r} P_j \right) \right] = D.$$

Summing (9.10) proves $\text{tr} C = Q^2 D$. The same identity and (9.11) give

$$\text{tr} B = QD, \quad \mathbf{1}^* B \mathbf{1} = D \sum_{q,r \in \mathcal{Q}} c_{qr}^2 \geq \frac{DQ^2}{K}. \quad (9.12)$$

To see the inequality, set $M = \sum_{q \in \mathcal{Q}} u_q u_q^\top$, a positive semidefinite K -by- K real matrix. Then $\text{tr} M = Q$ and $\text{tr}(M^2) = \sum_{q,r} c_{qr}^2$. Cauchy–Schwarz applied to its K nonnegative eigenvalues yields $\text{tr}(M^2) \geq (\text{tr} M)^2 / K = Q^2 / K$. Equivalently, the vector Gram matrix has rank at most K .

The support bound forced by separability. Suppose that C is separable over $\mathbb{C}^n \otimes \mathbb{C}^n$. Spectrally decomposing its local positive factors and absorbing positive weights into the vectors gives a finite decomposition

$$C = \sum_{\ell} (x^{(\ell)} \otimes y^{(\ell)})(x^{(\ell)} \otimes y^{(\ell)})^*, \quad x^{(\ell)}, y^{(\ell)} \in \mathbb{C}^n.$$

Whenever $\text{tr}(P_i P_j) = 0$, the corresponding tensor-basis diagonal entry is

$$0 = C_{(i,j),(i,j)} = \sum_{\ell} |x_i^{(\ell)} y_j^{(\ell)}|^2.$$

Each summand is nonnegative, so $x_i^{(\ell)} y_j^{(\ell)} = 0$ for every ℓ . For one summand put $z_i = x_i y_i$. If distinct i, j lie in $\text{supp } z$, then $x_i y_j \neq 0$. The preceding implication and (9.1) force $\text{tr}(P_i P_j) > 0$. The second hypothesis therefore gives $|\text{supp } z| \leq \kappa$.

Compressing the separable decomposition to the diagonal tensor subspace gives $B = \sum_{\ell} z^{(\ell)} (z^{(\ell)})^*$. Complex Cauchy–Schwarz on each support now yields

$$\begin{aligned} \mathbf{1}^* B \mathbf{1} &= \sum_{\ell} \left| \sum_i z_i^{(\ell)} \right|^2 \leq \kappa \sum_{\ell} \|z^{(\ell)}\|^2 = \kappa \operatorname{tr} B \\ &< \frac{Q}{K} \operatorname{tr} B = \frac{DQ^2}{K}. \end{aligned} \quad (9.13)$$

The inequality is also valid for a zero $z^{(\ell)}$, whose support is empty. The strict step uses $\kappa < Q/K$ and $\operatorname{tr} B = QD > 0$. This contradicts (9.12) and proves nonseparability. $\square$

The complex criterion has a state interpretation without the reality assumption. By (9.8), Lemma 2.6 gives

$$C = ((\Psi^\dagger \circ T_D) \otimes \Psi^\dagger)(\Omega_D \Omega_D^*). \quad (9.14)$$

Both local maps in this formula satisfy the input-transpose CP conditions, as the transfer lemma asserts for the PPT pair $\Psi, \Psi^\dagger$. Thus $C/(Q^2 D) \in \mathcal{C}$ also in the complex case, and Theorem 5.3 and Proposition 5.4 apply to it. The two local maps in (9.14) may differ; the reality assumption is what gives the same channel in (9.16).

Corollary 9.2 (The same local channel for real projections). *Under the hypotheses of Proposition 9.1, suppose that every P_i is real symmetric in the fixed basis of $\mathbb{C}^D$, and put $\Lambda = \Psi^\dagger : M_D(\mathbb{C}) \rightarrow M_n(\mathbb{C})$. Then, on all complex inputs,*

$$\Lambda \circ T_D = \Lambda = T_n \circ \Lambda, \quad \operatorname{tr} \Lambda(A) = Q \operatorname{tr} A. \quad (9.15)$$

In particular, Λ/Q is a trace-preserving PPT channel. For $\Phi_D = \Omega_D/\sqrt{D}$, the normalized maximally entangled vector on $\mathbb{C}^D \otimes \mathbb{C}^D$, one has

$$\frac{C}{Q^2 D} = \left(\frac{\Lambda}{Q} \otimes \frac{\Lambda}{Q} \right) (\Phi_D \Phi_D^*). \quad (9.16)$$

This is an entangled density operator on $\mathbb{C}^n \otimes \mathbb{C}^n$ in $\mathcal{C}$.

Proof. Reality and (9.5) give the additional identity $H_{ij}^\top = H_{ij}$. Equation (9.7) therefore becomes

$$[\Lambda(A)]_{ij} = \operatorname{tr}(H_{ij} A), \quad [\Lambda(T_D(A))]_{ij} = \operatorname{tr}(H_{ij}^\top A) = \operatorname{tr}(H_{ij} A).$$

This proves the first transpose invariance; the second is (9.8). Also,

$$\operatorname{tr} \Lambda(A) = \sum_i \operatorname{tr}(H_{ii} A) = \sum_{q \in \mathcal{Q}} \sum_{i \in \mathcal{I}_q} \operatorname{tr}(P_i A) = Q \operatorname{tr} A.$$

Since Λ is CP and PPT by the proposition, Λ/Q is a trace-preserving PPT channel with input size D and output size n .

Apply Lemma 2.6 to the pair Ψ, Λ . Its first local factor is

$$\Psi^\sharp = T_n \circ \Lambda \circ T_D = \Lambda$$

by the two invariances. Thus

$$G := (\Lambda \otimes \Lambda)(\Omega_D \Omega_D^*) = C. \quad (9.17)$$

Dividing this identity by $Q^2 D$ gives (9.16). Proposition 9.1 gives $\operatorname{tr} C = Q^2 D > 0$ and complex nonseparability. Finally, Λ/Q and $(\Lambda/Q) \circ T_D$ are CP, so (9.16) is a representation of the form in Definition 2.4. This proves the assertion. $\square$

The two channel uses in (9.16) act on separate subsystems. We next embed the rectangular maps into one matrix algebra to obtain a composition of square PPT maps.

Corollary 9.3 (Square embedding). *Under the hypotheses of Proposition 9.1, suppose in addition that $n \geq D$, and let $U : \mathbb{C}^D \rightarrow \mathbb{C}^n$ be the real coordinate isometry onto the first D coordinates. Define*

$$\Xi_1 = \text{Ad}_U \circ \Psi, \quad \Xi_2 = \Psi^\dagger \circ \text{Ad}_{U^*} = \Xi_1^\dagger. \quad (9.18)$$

Then $\Xi_1, \Xi_2 : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$ are PPT and

$$\Xi_2 \circ \Xi_1 = \Psi^\dagger \circ \Psi.$$

Their composition is not entanglement breaking.

Proof. Embedding and compression by U are CP. The PPT composition stability in Lemma 2.1 therefore applies to both maps. Directly, the reality of U also gives $T_n(UAU^*) = UT_D(A)U^*$, while (9.8) fixes the output transpose of the second map. The adjoint identity follows by reversing the two factors, and $U^*U = I_D$ gives the composite identity. Its Choi matrix is the nonseparable C of Proposition 9.1; the characterization in Lemma 2.2 proves the last assertion. $\square$

The dimension comparison in this corollary is used only to choose the isometry U . The rectangular criterion itself has no such comparison. The displayed square maps need not preserve trace or be unital.

10 A finite combinatorial realization

We construct data satisfying Proposition 9.1. A restricted-intersection bound gives real unit vectors such that every sufficiently large subfamily contains an orthogonal pair. The fiber argument for threefold tensor products strengthens this conclusion to eight mutually orthogonal members in every sufficiently large subfamily. Six of these eight vectors will index the six measurements in a parity obstruction; the remaining two receive a trivial measurement. Tensoring these measurements over the eight-subsets gives the projections required by the criterion. Because the projections are real, the resulting state is produced by the same trace-preserving PPT channel on both halves of a maximally entangled state.

10.1 Vectors with forced orthogonality

Set $p = 257$, $k = 4p = 1028$, and $[k] = \{1, \dots, k\}$. Define

$$\begin{aligned} \mathcal{U} &= \{A \subseteq [k] : |A| = 2p, 1 \in A\}, \\ b &= |\mathcal{U}| = \binom{1027}{513} = \frac{1}{2} \binom{4p}{2p}, \\ s_A &= \frac{1}{\sqrt{k}} (2\mathbf{1}_{\{h \in A\}} - 1)_{h=1}^k \in \mathbb{R}^k \quad (A \in \mathcal{U}). \end{aligned} \quad (10.1)$$

Each s_A is a unit vector. Since $|A \triangle B| = 4p - 2|A \cap B|$, counting agreeing and disagreeing coordinates gives

$$s_A^\top s_B = \frac{k - 2|A \triangle B|}{k} = \frac{|A \cap B|}{p} - 1. \quad (10.2)$$

Thus two base vectors are orthogonal exactly when their sets intersect in p elements. Put

$$a_0 = \sum_{j=0}^{p-1} \binom{k}{j} = \sum_{j=0}^{256} \binom{1028}{j}, \quad \delta = \frac{a_0}{b}. \quad (10.3)$$

Frankl and Rödl's theory of forbidden intersections also forces orthogonal families of a prescribed size in large subsets of a sign cube [22, Theorem 1.11]. Here the precise bound follows from a direct polynomial argument and the fiber amplification below.

Lemma 10.1 (Restricted intersections). *If $\mathcal{F} \subseteq \mathcal{U}$ contains no two distinct sets whose associated vectors are orthogonal, then $|\mathcal{F}| \leq a_0 = \delta b$.*

Proof. The restricted-intersection estimate belongs to the Frankl–Wilson theory [23]. The proof below uses the multilinear-polynomial approach of Alon, Babai, and Suzuki [2]. We give the needed specialization in full. The integer 257 is prime: the primes at most $\sqrt{257}$ are 2, 3, 5, 7, 11, 13, and the respective remainders are 1, 2, 2, 5, 4, 10.

For distinct $A, B \in \mathcal{F}$, their common element 1 and their equal sizes give $1 \leq |A \cap B| \leq 2p - 1$. Equation (10.2) excludes the value p . Hence $|A \cap B|$ is nonzero modulo p , whereas $|A \cap A| = 2p$ is zero modulo p . Over the field $\mathbb{F}_p$, define

$$f_A(x_1, \dots, x_k) = 1 - \left(\sum_{h \in A} x_h \right)^{p-1} \quad (A \in \mathcal{F}).$$

For every nonzero $a \in \mathbb{F}_p$, multiplication by a permutes the nonzero field elements; comparing their products gives $a^{p-1} = 1$. Therefore, on the indicator vectors of the members of $\mathcal{F}$,

$$f_A(\mathbf{1}_B) = \begin{cases} 1, & A = B, \\ 0, & A \neq B. \end{cases}$$

Replace every positive power of each variable in every monomial by its first power. This preserves all evaluations on Boolean vectors and leaves degree at most $p - 1$. The resulting multilinear polynomials remain linearly independent: evaluating a linear relation at each $\mathbf{1}_B$ makes every coefficient zero. The space of multilinear polynomials of degree at most $p - 1$ has a basis consisting of the a_0 squarefree monomials of those degrees. Thus $|\mathcal{F}| \leq a_0$. $\square$

Lemma 10.2 (Fiber amplification). *For $t = 1, 2, 3$, associate to $(A_1, \dots, A_t) \in \mathcal{U}^t$ the unit vector $s_{A_1} \otimes \dots \otimes s_{A_t}$. If $\mathcal{R} \subseteq \mathcal{U}^t$ contains no collection of 2^t distinct tuples whose associated vectors are pairwise orthogonal, then*

$$|\mathcal{R}| \leq t\delta b^t.$$

Proof. The case $t = 1$ is Lemma 10.1. For $t = 2$ or 3, write

$$\mathcal{R}_A = \{(A_2, \dots, A_t) : (A, A_2, \dots, A_t) \in \mathcal{R}\}, \quad A \in \mathcal{U}.$$

Let $\mathcal{G}$ consist of those A for which $\mathcal{R}_A$ contains 2^{t-1} tuples with pairwise orthogonal tensor vectors in the last $t - 1$ factors. If distinct $A, B \in \mathcal{G}$ had $s_A \perp s_B$, choose one such collection in each fiber. Within either lifted collection, orthogonality comes from the last factors; between the two collections, it comes from the first factors. The two fibers are disjoint, so this would produce 2^t distinct tuples forbidden by the hypothesis. Thus $\mathcal{G}$ has no orthogonal base pair, and $|\mathcal{G}| \leq a_0 = \delta b$ by the preceding lemma.

Each fiber in $\mathcal{G}$ has at most b^{t-1} elements. Each other fiber has at most $(t-1)\delta b^{t-1}$ elements by the induction hypothesis. Summing these bounds and using at most b fibers of the latter kind gives

$$|\mathcal{R}| \leq (\delta b)b^{t-1} + b(t-1)\delta b^{t-1} = t\delta b^t.$$

□

Use triples as measurement indices, called *questions*:

$$\begin{aligned} \mathcal{Q} = \mathcal{U}^3, \quad Q = b^3, \quad K = k^3 = 1028^3 = 1086373952, \\ u_q = s_{A_1} \otimes s_{A_2} \otimes s_{A_3} \in \mathbb{R}^K \quad (q = (A_1, A_2, A_3)). \end{aligned} \quad (10.4)$$

The scalar target is $3\delta < 1/K$: the support bound $3\delta Q$ must be strictly below Q/K for the projection criterion. These vectors are unit vectors. If $q = (A_1, A_2, A_3)$ and $r = (B_1, B_2, B_3)$, their inner product is the rational number

$$c_{qr} = u_q^\top u_r = \prod_{h=1}^3 \left(\frac{|A_h \cap B_h|}{257} - 1 \right). \quad (10.5)$$

An *orthogonal eight-set* will mean an eight-element subset of $\mathcal{Q}$ whose associated vectors are pairwise orthogonal. By Lemma 10.2, with $t = 3$, every $\mathcal{R} \subseteq \mathcal{Q}$ containing no orthogonal eight-set satisfies

$$|\mathcal{R}| \leq \kappa, \quad \kappa := 3\delta Q = 3a_0 b^2. \quad (10.6)$$

The next analytic estimate is what makes this bound strict enough for Proposition 9.1. The central coefficient is largest among the $4p+1$ coefficients of $(1+1)^{4p}$, so

$$b = \frac{1}{2} \binom{4p}{2p} \geq \frac{16^p}{2(4p+1)}.$$

The binomial coefficients increase up to the central one, and the degree- p term in the expansion of $(1+1/3)^{4p}$ is no greater than the full sum. Hence

$$a_0 \leq p \binom{4p}{p} \leq p(4/3)^{4p} 3^p = p(256/27)^p < p 12^p.$$

It follows that

$$\delta < 2p(4p+1)(3/4)^p < 2^{-65}. \quad (10.7)$$

For the last inequality, use $2p(4p+1) = 528906 < 2^{20}$, $p = 3 \cdot 85 + 2$, and $(3/4)^3 = 27/64 < 1/2$; the remaining factor $(3/4)^2$ is less than one. Also $1028 < 2^{11}$, so

$$K < 2^{33}, \quad 3\delta < 3 \cdot 2^{-65} < 2^{-33} < \frac{1}{K}.$$

Combining this with (10.6) proves

$$\kappa = 3a_0 b^2 < \frac{Q}{K}. \quad (10.8)$$

The proof has now supplied the vector and cardinality ingredients. We next construct measurements that turn every orthogonal eight-set into a forbidden positive-overlap choice.

10.2 Six parity measurements

On $\mathbb{C}^2$, let

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Use the Mermin–Peres square [42, 47, 48]; its interpretation in the setting of quantum magic games is discussed by Arkhipov [4]. The margins in the following array give the products of the three observables in each row and column:

$$\begin{array}{ccc|c} X \otimes I_2 & I_2 \otimes X & X \otimes X & +I_4 \\ I_2 \otimes Z & Z \otimes I_2 & Z \otimes Z & +I_4 \\ X \otimes Z & Z \otimes X & Y \otimes Y & +I_4 \\ \hline +I_4 & +I_4 & -I_4 & \end{array} \quad (10.9)$$

All nine entries are real symmetric involutions. In particular, if $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, then $Y = iJ$ and $Y \otimes Y = -J \otimes J$, a real integral symmetric matrix. Order the six contexts as the three rows followed by the three columns, each read left to right or top to bottom. In context t , write the three observables as (O_1, O_2, O_3) . The first two commute and

$$O_3 = e_t O_1 O_2, \quad (e_1, e_2, e_3, e_4, e_5, e_6) = (1, 1, 1, 1, 1, -1). \quad (10.10)$$

These identities follow from $XZ = -iY$ and $ZX = iY$. For example, the first two entries of the last row multiply, in either order, to $Y \otimes Y$; those of the last column multiply, in either order, to $-Y \otimes Y$. Since O_3 is a signed product of the first two, all three commute.

Let $\mathcal{A} = \{-1, +1\}^2$. The four outcomes of context t are the matrices

$$h_t(\alpha, \beta) = \frac{1}{4}(I_4 + \alpha O_1)(I_4 + \beta O_2), \quad (\alpha, \beta) \in \mathcal{A}. \quad (10.11)$$

The two factors divided by two are commuting spectral projections. Their products are therefore self-adjoint projections. Distinct outcomes are orthogonal, and the four products sum to I_4 . They are real symmetric with rational entries, and their ranges have the three context eigensigns $(\alpha, \beta, e_t \alpha \beta)$.

Lemma 10.3 (Parity obstruction). *There is no choice of one projection from each of the six measurements in (10.11) such that the trace overlap of every two chosen projections from distinct contexts is strictly positive.*

Proof. Suppose there were such a choice. All six selected projections would be nonzero. Let a selected row projection E and column projection F assign eigensigns $a, b \in \{-1, +1\}$ to their shared observable O . The eigenspace identities and self-adjointness give

$$aEF = EOF = bEF.$$

If $a \neq b$, then $EF = 0$ and their overlap is zero, contrary to the hypothesis. Thus the row and column assignments agree at each of the nine entries of the square.

The choice would consequently give nine consistent signs. The product of the three signs in context t is e_t , by (10.10). Multiplying the three row constraints gives product $+1$ for the nine signs; multiplying the three column constraints gives product -1 . This is a contradiction. The argument applies to eigenspaces in the complex Hilbert space $\mathbb{C}^4$. $\square$

10.3 Tensor assembly and consequences

We now impose the six-context obstruction on every orthogonal eight-set of questions. Each such set receives its own tensor factor. Six of its questions use the six parity measurements on that factor, while all other questions use the trivial measurement with one outcome I_4 and three zero outcomes. Thus any positive-overlap choice containing the eight-set violates parity on its factor. This arrangement also ensures the commutation needed in the projection criterion: two distinct questions with nonzero vector inner product cannot both have nontrivial measurements on any one factor.

For a uniform index set, include a factor carrying only trivial measurements for every nonorthogonal eight-subset as well. Let

$$\begin{aligned} \mathcal{S} &= \{S \subseteq \mathcal{Q} : |S| = 8\}, & L &= |\mathcal{S}| = \binom{Q}{8}, \\ \mathcal{H} &= \bigotimes_{S \in \mathcal{S}} \mathbb{C}^4 \cong \mathbb{C}^D, & D &= 4^L. \end{aligned} \tag{10.12}$$

To fix all bases, order subsets of $[k]$ by their increasing lists, then order tuples lexicographically, and order the eight-subsets by their increasing lists in that order. Use the corresponding order for the tensor factors, the standard tensor-product basis in each $\mathbb{C}^4 = \mathbb{C}^2 \otimes \mathbb{C}^2$, and the lexicographic order on sign pairs with $-1 < +1$.

For $S \in \mathcal{S}$ and $q \in \mathcal{Q}$, define a four-outcome measurement $h_{S,q}$ on the factor indexed by S . If S is an orthogonal eight-set and q is its t -th element with $1 \leq t \leq 6$, set $h_{S,q} = h_t$. In every other case, set

$$h_{S,q}(+1, +1) = I_4, \quad h_{S,q}(\alpha, \beta) = 0 \quad \text{when } (\alpha, \beta) \neq (+1, +1). \tag{10.13}$$

Thus the last two questions in an orthogonal eight-set use the trivial measurement, as do all questions outside that set and all questions on a nonorthogonal factor. Every decision of whether a factor is orthogonal is determined by the rational numbers in (10.5).

For each q , retain an outcome string on every factor and define

$$\mathcal{I}_q = \{q\} \times \mathcal{A}^{\mathcal{S}}, \quad P_{q,\lambda} = \bigotimes_{S \in \mathcal{S}} h_{S,q}(\lambda_S) \quad (\lambda \in \mathcal{A}^{\mathcal{S}}). \tag{10.14}$$

Order these outcome strings lexicographically using the stated factor and sign-pair orders, and order the full labels first by question and then by outcome string. This is the standard basis order used for the concrete maps. The matrices $P_{q,\lambda}$ are real symmetric projections. For a fixed q , two distinct strings differ in some factor, where their local projections have zero product; hence their global projections are orthogonal. Summing over all strings factors as $\bigotimes_S \sum_{\alpha, \beta} h_{S,q}(\alpha, \beta) = I_D$. Keeping every outcome string, including those whose projection is zero, gives a uniform label set: there are exactly $4^L = D$ labels for every question. Hence

$$m := \left| \bigsqcup_{q \in \mathcal{Q}} \mathcal{I}_q \right| = QD. \tag{10.15}$$

In particular $m \geq D$, as required for the square embedding in Corollary 9.3.

Lemma 10.4. *The vectors (10.4) and measurements (10.14) satisfy the two hypotheses of Proposition 9.1 with $n = m = QD$ and $\kappa = 3a_0b^2 < Q/K$.*

Proof. If $q = r$, projections with those labels belong to one projective measurement and commute. Suppose that $q \neq r$ and $c_{qr} \neq 0$. They cannot both receive nontrivial contexts on any one factor: that would place them together in an orthogonal eight-set, forcing $c_{qr} = 0$. On each factor at least one of the two selected local projections is consequently 0 or I_4 . They commute on each factor, so their tensor products commute. This proves the first hypothesis.

Now let J be a label set with strictly positive trace overlap between every distinct pair. Distinct labels for one question have zero product, so J contains at most one label per question. The trace overlap of any two global projections factors as

$$\text{tr}(P_{q,\lambda} P_{r,\mu}) = \prod_{S' \in \mathcal{S}} \text{tr}(h_{S',q}(\lambda_{S'}) h_{S',r}(\mu_{S'})). \quad (10.16)$$

Every local factor is nonnegative by (9.1), whether or not the two local projections commute. Thus strict positivity of this finite product implies strict positivity of each factor.

If the questions represented by J contained an orthogonal eight-set S , take the selected labels for its first six questions. At the factor indexed by S , equation (10.16) would give strictly positive overlap for each pair of the six selected context projections. Lemma 10.3 forbids this. The represented questions therefore contain no orthogonal eight-set, and (10.6) gives $|J| \leq \kappa$. For completeness, the pairwise condition on a singleton is vacuous even when its projection is zero; it still satisfies the bound because $a_0, b \geq 1$ and $\kappa = 3a_0b^2 \geq 3$. The strict inequality $\kappa < Q/K$ is (10.8). $\square$

Theorem 10.5 (Finite combinatorial construction). *Set*

$$\begin{aligned} p &= 257, & k &= 1028, & b &= \binom{1027}{513}, & a_0 &= \sum_{j=0}^{256} \binom{1028}{j}, \\ Q &= b^3, & K &= 1086373952, & \kappa &= 3a_0b^2 < \frac{Q}{K}, \\ L &= \binom{Q}{8}, & D &= 4^L, & m &= QD. \end{aligned} \quad (10.17)$$

Use the vectors in (10.4), the D labels per question and projections in (10.14), and the map $\Psi : M_m(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ in (9.2). Put $\Lambda = \Psi^\dagger$ and $\Phi_D = \Omega_D/\sqrt{D}$.

1. The map $\Lambda/Q : M_D(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ is a trace-preserving PPT channel, and

$$\rho_{\text{comb}} = \left(\frac{\Lambda}{Q} \otimes \frac{\Lambda}{Q} \right) (\Phi_D \Phi_D^*) = \frac{J(\Psi^\dagger \circ \Psi)}{Q^2 D} \quad (10.18)$$

is an entangled density operator on $\mathbb{C}^m \otimes \mathbb{C}^m$ belonging to $\mathcal{C}$. It has $K_D(\rho_{\text{comb}}) = 0$ and satisfies the uniform unhalved $1/5$ gap and all operational consequences of Theorem 5.3 and Proposition 5.4, with their stated protocol, purification, and completion hypotheses.

2. For the real coordinate isometry $U : \mathbb{C}^D \rightarrow \mathbb{C}^m$ onto the first D coordinates, the maps

$$\Phi_1^{\text{comb}} = \text{Ad}_U \circ \Psi, \quad \Phi_2^{\text{comb}} = \Lambda \circ \text{Ad}_{U^*} = (\Phi_1^{\text{comb}})^\dagger$$

are PPT maps on $M_m(\mathbb{C})$, and

$$J(\Phi_2^{\text{comb}} \circ \Phi_1^{\text{comb}}) = Q^2 D \rho_{\text{comb}} \quad (10.19)$$

is nonseparable over $\mathbb{C}^m \otimes \mathbb{C}^m$. In particular their composition is not entanglement breaking.

3. Let Θ_{comb} be the channel obtained by applying Proposition 8.1 to this square pair. It is a trace-preserving PPT channel on $M_{2m+1}(\mathbb{C})$. With the positive constants c_1, c_2 and the first coordinate inclusion W_0 from that proposition, the Choi matrix of its square has the corner

$$(W_0^* \otimes W_0^*)J(\Theta_{\text{comb}}^2)(W_0 \otimes W_0) = c_1 c_2 Q^2 D \rho_{\text{comb}}. \quad (10.20)$$

Consequently $\Theta_{\text{comb}}^2 = \Theta_{\text{comb}} \circ \Theta_{\text{comb}}$ is not entanglement breaking.

Proof. Lemma 10.4 verifies the hypotheses of Proposition 9.1. Its composite $C = J(\Psi^\dagger \circ \Psi)$ is positive and nonseparable over $\mathbb{C}^m \otimes \mathbb{C}^m$, with $\text{tr } C = Q^2 D$. The projections are real symmetric, so Corollary 9.2 proves (10.18), the channel assertion, and membership in $\mathcal{C}$. Theorem 5.3 and Proposition 5.4 give the stated operational conclusions.

Since $m = QD \geq D$, Corollary 9.3 applies to the displayed coordinate isometry and gives the square PPT pair. Its composite is $\Psi^\dagger \circ \Psi$, and the real identification (9.17) gives (10.19). Applying Proposition 8.1 to this pair gives the channel on $M_{2m+1}(\mathbb{C})$ and its exact corner (10.20). The corner is nonseparable, so that proposition also proves that the channel square is not entanglement breaking. $\square$

All these objects are given by finite exact formulas. The normalized sign vectors s_A have coordinates $\pm 1/\sqrt{1028} = \pm 1/(2\sqrt{257})$, which are irrational because 257 is prime. The Gram entries of the question vectors in (10.5), rather than the vector coordinates, enter the Choi blocks. Those Gram entries are rational; all nine local observables are integral, and all local and global projections are rational. Consequently every block H_{ij} , every coefficient of $\Psi, \Lambda, \Phi_1^{\text{comb}}, \Phi_2^{\text{comb}}$, and every entry of ρ_{comb} is rational in the specified bases. The completion also has rational coefficients: its positive scalings and flag matrices in Proposition 8.1 are formed from the rational square maps using traces and rational operations. Appendix B describes supplementary exact checks of the small arithmetic and six-context calculations.

A Details of the dimension-ten certificate

A.1 Finite arithmetic

This subsection gives the exact arithmetic support for the block identities (7.4) and the finite-field data in Section 7. The Python 3 program below uses only the standard library and integer arithmetic. Its input matrix entries are those of (7.1), with zero-based row indices in the code.

The function `checksyz` uses only $i < j$ and column indices $a < b$. The cases $i = j$ and $a = b$ vanish identically, while exchanging either pair of indices changes the sign of the same difference. Thus these strict loops check every equality in (7.4).

The code variable `R` is the coefficient array $[D \mid H]$, and `T` is the reduction of N modulo the current prime. The program expands each determinant over the $4!$ permutations, collects coefficients in the order (7.5), and performs Gaussian elimination on H modulo each prime. The columns of `T` are the reductions of the columns in (7.7).

For $f(z) = z^{20} + c_1 z^{19} + \cdots + c_{20}$, with $c_0 = 1$, the program uses the Newton identities

$$k c_k = - \sum_{i=0}^{k-1} c_i \text{tr}(N^{k-i}) \quad (1 \leq k \leq 20).$$

Division by k is valid at each of 41, 131, 139. Repeated p th powers modulo $\bar{f}$ then give the residues of z^{p^k} , and the polynomial Euclidean algorithm gives $d_p(k)$. All inverses are taken in the stated prime field. No floating-point approximation or heuristic search is involved.

The program asserts the integer block identities and prints the modular data. Verification of the modular certificate includes comparing the printed output with the recorded values below. The operator, projective, and Galois deductions are proved in Sections 6 and 7, using the algebraic facts proved in the next subsection. The program does not perform a general elimination or Galois-group computation.

From the paper directory, run `python3 verification/exact_certificate.py` with assertions enabled, without the `-O` or `-OO` optimization options, and with `PYTHONOPTIMIZE` unset.

The listing uses an identical build copy of the verification source; its line numbers refer to that source file.

Listing 1: Exact verification of the matrix identities and modular certificate.

```

1 from itertools import combinations, permutations
2
3 M = [
4     [[6,0,0,0],[0,6,0,0],[0,0,6,0],[0,0,0,6],[0,0,0,0],[0,0,0,0]],
5     [[-12,0,0,0],[0,-6,0,0],[0,0,6,0],[0,0,0,12],[-6,0,6,6],[-6,-6,6,0]],
6     [[0,6,-2,0],[6,6,0,2],[-2,0,10,0],[0,2,0,24],[0,0,0,6],[0,-6,6,-6]],
7     [[0,0,-4,-3],[0,-2,-3,-4],[-4,-3,11,6],[-3,-4,6,25],[6,6,0,0],[0,-6,6,6]]
8 ]
9
10 def checksym():
11     for i,j in combinations(range(4),2):
12         for a,b in combinations(range(4),2):
13             assert sum(M[i][r][a]*M[j][r][b]-M[j][r][a]*M[i][r][b] for r in range(6))==0
14
15 mons=[(a,b,d-a-b) for d in range(5) for a in range(d+1) for b in range(d-a+1)]
16 idx={u:i for i,u in enumerate(mons)}
17 eps=[(0,0,0),(1,0,0),(0,1,0),(0,0,1)]
18 def add(u,v):
19     return tuple(a+b for a,b in zip(u,v))
20 def coefficients():
21     R=[]
22     for rows in combinations(range(6),4):
23         coeff=[0]*35
24         for perm in permutations(range(4)):
25             sg=(-1)**sum(perm[i]>perm[j] for i,j in combinations(range(4),2))
26             D={(0,0,0):sg}
27             for r,c in zip(rows,perm):
28                 E={}
29                 for k in range(4):
30                     if M[k][r][c]:
31                         for u,v in D.items():
32                             t=add(u,eps[k])
33                             E[t]=E.get(t,0)+v*M[k][r][c]
34             D=E
35             for u,v in D.items(): coeff[idx[u]]+=v
36     R.append(coeff)
37     return R
38
39 def mult1(R,p):
40     B=[[v%p for v in row] for row in R]
41     detH=1
42     for i in range(15):
43         piv=next(j for j in range(i,15) if B[j][20+i])
44         if piv!=i:
45             B[i],B[piv]=B[piv],B[i]
46             detH=(-detH)%p

```

```

47     c=B[i][20+i]; detH=detH*c%p
48     inv=pow(c,p-2,p)
49     B[i]=[(x*inv)%p for x in B[i]]
50     for j in range(15):
51         if j!=i:
52             c=B[j][20+i]
53             B[j]=[(x-c*y)%p for x,y in zip(B[j],B[i])]
54     T=[0]*20 for _ in range(20)]
55     for j,u in enumerate(mons[:20]):
56         k=idx[add(u,(1,0,0))]
57         if k<20: T[k][j]=1
58         else:
59             for i in range(20): T[i][j]=-B[k-20][i]%p
60     return detH,T
61 def mm(A,B,p):
62     BT=list(zip(*B))
63     return [[sum(x*y for x,y in zip(row,col))%p for col in BT] for row in A]
64 def char(T,p):
65     power=[[int(i==j) for j in range(20)] for i in range(20)]
66     c=[1]
67     tr=[0]
68     for k in range(1,21):
69         power=mm(power,T,p)
70         tr.append(sum(power[i][i] for i in range(20))%p)
71         c.append((-pow(k,p-2,p)*sum(c[i]*tr[k-i] for i in range(k))%p)
72     return c
73
74 # lists for polynomials below are low degree first, mod p
75 def trim(f):
76     while f and f[-1]==0: f.pop()
77     return f
78 def sub(f,g,p):
79     h=f[:] + [0]*max(0,len(g)-len(f))
80     for i,x in enumerate(g): h[i]=(h[i]-x)%p
81     return trim(h)
82 def mul(f,g,p):
83     if not f or not g: return []
84     h=[0]*(len(f)+len(g)-1)
85     for i,x in enumerate(f):
86         for j,y in enumerate(g): h[i+j]=(h[i+j]+x*y)%p
87     return trim(h)
88 def rem(f,g,p):
89     h=f[:]; inv=pow(g[-1],p-2,p)
90     while len(h)>len(g):
91         k=len(h)-len(g); a=h[-1]*inv%p
92         for j,y in enumerate(g): h[k+j]=(h[k+j]-a*y)%p
93         trim(h)
94     return h
95 def gcd(f,g,p):
96     while g: f,g=g,rem(f,g,p)
97     return f
98 def pth(u,f,p):
99     n=p; a=u; b=[1]
100    while n:
101        if n%2: b=rem(mul(b,a,p),f,p)
102        a=rem(mul(a,a,p),f,p); n//=2
103    return b
104 def gcddegrees(coef,p):
105     f=coef[::-1]

```

```

106 z=[0,1]
107 u=z
108 ds=[]
109 for k in range(1,21):
110     u=pth(u,f,p)
111     ds.append(len(gcd(f,sub(u,z,p),p))-1)
112 return ds
113
114 checksym()
115 R=coefficients()
116 for p in (41,131,139):
117     detH,T=mult1(R,p)
118     c=char(T,p)
119     print(p,detH)
120     print(c)
121     print(gcddegrees(c,p))

```

The file `verification/certificate-output.txt` records the values below; an identical build copy is used for typesetting. Coefficients of $\bar{f}$ are in leading-first order, and the gcd-degree lists are ordered by $k = 1, \dots, 20$. The first line for each prime gives p and $\det H \bmod p$.

Listing 2: Recorded output of the exact verification program.

```

41 40
[1, 6, 22, 32, 32, 4, 37, 27, 19, 9, 26, 6, 32, 4, 4, 4, 33, 2, 12, 24, 36]
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 20]
131 23
[1, 100, 35, 54, 51, 41, 36, 76, 115, 33, 118, 25, 65, 61, 124, 110, 18, 61, 53, 39, 17]
[1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 18, 3, 1, 3]
139 9
[1, 40, 61, 0, 51, 49, 50, 50, 107, 89, 57, 84, 54, 20, 40, 37, 60, 126, 44, 44, 97]
[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 20, 1]

```

A.2 Algebraic facts

We prove the three general lemmas used in Section 7. The first supplies a common zero used to prove the affine quotient nonzero; the second accommodates rational coefficients with denominators prime to p ; the third converts squarefree modular factor degrees into Galois permutations.

Proof of Lemma 7.3. Let $X_0, \dots, X_3, Y_0, \dots, Y_3$ be indeterminates, put $u_{ij} = X_i Y_j$, and consider the finitely generated complex algebra

$$\mathcal{R} = \mathbb{C}[u_{ij} : 0 \leq i, j < 4] \subset \mathbb{C}[X_0, \dots, X_3, Y_0, \dots, Y_3].$$

It is a Noetherian domain. Its closed complex points are exactly the matrices of rank at most one. Indeed, the identities $u_{ij} u_{k\ell} = u_{i\ell} u_{kj}$ force the values at any point to have rank at most one, and every such matrix is $xy^\top$ and defines an evaluation of $\mathcal{R}$. The maximal ideal $\mathfrak{m} = (u_{ij} : 0 \leq i, j < 4)$ corresponds to the zero matrix.

For $1 \leq a, b \leq 4$, put $P_{a,b} = (u_{ij} : i \geq a \text{ or } j \geq b)$. The quotient by $P_{a,b}$ is the domain $\mathbb{C}[X_i Y_j : 0 \leq i < a, 0 \leq j < b]$. To justify the identification, use the monomial basis $X^\alpha Y^\beta$ with $|\alpha| = |\beta|$ for $\mathcal{R}$: a monomial containing an excluded variable factors by an excluded generator, while the remaining monomials stay distinct. Hence

$$0 = P_{4,4} \subsetneq P_{3,4} \subsetneq P_{2,4} \subsetneq P_{1,4} \subsetneq P_{1,3} \subsetneq P_{1,2} \subsetneq P_{1,1} \subsetneq \mathfrak{m}$$

is a strict chain of prime ideals. In particular, $\text{ht } \mathfrak{m} \geq 7$.

Let $\ell_1, \dots, \ell_6 \in \mathcal{R}$ be the linear combinations of the u_{ij} representing the six bilinear forms, and put $\mathfrak{a} = (\ell_1, \dots, \ell_6)$. If the only common zero on the rank-at-most-one matrices were the zero matrix, the Hilbert Nullstellensatz for the finite-type complex algebra $\mathcal{R}$ would give $\sqrt{\mathfrak{a}} = \mathfrak{m}$. The prime $\mathfrak{m}$ would then be minimal over an ideal generated by six elements. Krull's height theorem bounds the height of such a prime in a Noetherian ring by six, contradicting the chain above [58, Tags 00FV and 0BBZ(1)]. Thus a nonzero rank-one matrix $xy^\top$ annihilates all six forms. Both factors are nonzero. $\square$

Proof of Lemma 7.4. Normalize the valuation by $v_p(p) = 1$. In either pair its values on nonzero elements are integers, and $R_p/pR_p = \mathbb{F}_p$. For a nonzero polynomial $a(z) = \sum_i a_i z^i$ over $\mathbb{L}$, put $\nu(a) = \min_{a_i \neq 0} v_p(a_i)$. Scaling nonzero polynomials a, b by $p^{-\nu(a)}$ and $p^{-\nu(b)}$ makes their coefficients integral with least valuation zero. Their reductions are nonzero and have nonzero product in the domain $\mathbb{F}_p[z]$. Thus

$$\nu(ab) = \nu(a) + \nu(b).$$

For monic g, h one has $\nu(g), \nu(h) \leq 0$, whereas the monic $P \in R_p[z]$ has $\nu(P) = 0$. Additivity forces $\nu(g) = \nu(h) = 0$, so all coefficients of both factors are integral. Their leading coefficients remain one after reduction, which preserves their degrees; a monic constant factor is 1 and retains degree zero. Any factorization of P into two positive-degree polynomials over $\mathbb{L}$ can be normalized to monic factors, whose reductions would still have positive degree. This contradicts irreducibility of the reduction of P . $\square$

Proof of Lemma 7.5. Embed the coefficients of F into $\mathbb{Z}_p$ and put $s = \text{lcm}(s_1, \dots, s_h)$. Choose a monic irreducible $\bar{g} \in \mathbb{F}_p[X]$ of degree s and a monic coefficient lift $g \in \mathbb{Z}_p[X]$; we use the standard finite-field fact that an irreducible polynomial of every positive degree exists. Lemma 7.4 for $(\mathbb{Q}_p, \mathbb{Z}_p)$ shows that g is irreducible over $\mathbb{Q}_p$. If α is a root, put

$$E = \mathbb{Q}_p(\alpha), \quad \mathcal{O} = \mathbb{Z}_p[\alpha] \simeq \mathbb{Z}_p[X]/(g).$$

The ring $\mathcal{O}$ is free of rank s over $\mathbb{Z}_p$, hence complete and separated for the p -adic topology, and $\mathcal{O}/p\mathcal{O} \simeq \mathbb{F}_{p^s}$. Every element with nonzero residue is a unit: lift an inverse modulo p and correct it by a convergent geometric series.

Every simple residue root of a polynomial in $\mathbb{Z}_p[z]$ lifts uniquely to a root in $\mathcal{O}$. To see this, suppose r_n is a root modulo p^n whose derivative has nonzero residue. The first-order Taylor congruence gives a unique residue class of c such that $r_{n+1} = r_n + p^n c$ is a root modulo p^{n+1} . Completeness gives a limiting root. If two roots have the same simple residue, the polynomial difference factorization writes their difference times a unit as zero; the roots therefore coincide.

Every root of $\bar{F}$ lies in $\mathbb{F}_{p^s}$ and is simple. Lifting them gives $\deg F$ distinct roots of F in $\mathcal{O}$, hence all the roots. Finite fields are perfect, so $\bar{\alpha}^p$ is a simple root of $\bar{g}$. Lift it to a root $\beta \in \mathcal{O}$ of g . Sending α to β defines a $\mathbb{Q}_p$ -embedding $E \rightarrow E$, because g is irreducible. This embedding is an automorphism σ , as E is finite-dimensional over $\mathbb{Q}_p$.

Since $\beta \in \mathcal{O}$, we have $\sigma(\mathcal{O}) \subseteq \mathcal{O}$. The induced residue map fixes $\mathbb{F}_p$ and sends $\bar{\alpha}$ to $\bar{\alpha}^p$, so it is Frobenius on $\mathbb{F}_{p^s}$. In a $\mathbb{Z}_p$ -basis of $\mathcal{O}$, the matrix of this endomorphism reduces to an invertible residue matrix. Its determinant is a p -adic unit, and therefore $\sigma(\mathcal{O}) = \mathcal{O}$.

If ρ_r is the lifted root of F with residue r , uniqueness of lifting gives $\sigma(\rho_r) = \rho_{r^p}$. Thus the permutation of the lifted roots has exactly the Frobenius cycle lengths on their residues, namely $s_1, \dots, s_h$. The subfield of E generated over $\mathbb{Q}$ by these roots is a rational splitting field of F . The automorphism σ permutes its generating roots and hence preserves this subfield. Its restriction

is the required rational Galois automorphism. A $\mathbb{Q}$ -isomorphism transports the statement to any chosen splitting field. $\square$

B Supplementary exact checks

The construction in Section 10 is proved by finite formulas and analytic inequalities. The four accompanying files in `verification/` preserve two exact integer checks of its small arithmetic components. These checks are supplementary: the restricted-intersection argument, fiber amplification, tensor construction, and nonseparability proof do not use their execution as a premise.

The program `combinatorics_exact_checks.py` uses only the Python standard library. It checks primality of 257 by trial division and verifies the intermediate binomial and power inequalities used in (10.7). It also checks $3a_0K < b$ and $3a_0b^2 \geq 3$. Its recorded output `combinatorics_exact_checks.json` stores the exact integers a_0 (under the key A) and b , confirms $K = 1086373952$, and records the stronger interval

$$2^{-194} \leq \delta < 2^{-193}. \quad (\text{B.1})$$

The analytic proof needs only $\delta < 2^{-65}$.

The program `gadget_exact_check.py` represents each local projection $h_t(\alpha, \beta)$ by the integral matrix $R = 4h_t(\alpha, \beta)$. It checks the observable relations, identities and eigensigns for all 24 projections, and measurement completeness, and computes the 36 ordered context-pair overlap tables. The stored integer $\text{tr}(RR')$ is exactly $16 \text{tr}(h_t h_{t'})$. It then enumerates all $4^6 = 4096$ selections of one outcome per context and counts zero overlaps among the 15 unordered pairs of distinct contexts. The recorded output `gadget_exact_check.json` has no selection with all overlaps positive and gives the following counts:

number of zero-overlap pairs	1	3	5	7	9
number of selections	144	1344	2016	576	16.

The files retain all exact matrices and overlap tables, not just this summary.

From the paper directory the outputs can be regenerated by

```
python3 verification/combinatorics_exact_checks.py
python3 verification/gadget_exact_check.py > verification/gadget_exact_check.json
```

Run these commands with Python assertions enabled: do not use `-0` or `-00`, and do not set `PYTHONOPTIMIZE`. All computations in these programs use exact Python integers and no floating-point tolerance. They do not instantiate the enormous global matrices or enumerate the question set; those objects are specified and analyzed by the finite formulas in the text.

References

- [1] A. Acín and N. Gisin, “Quantum correlations and secret bits,” *Physical Review Letters* **94**, 020501 (2005). doi:10.1103/PhysRevLett.94.020501. arXiv:quant-ph/0310054v3.
- [2] N. Alon, L. Babai, and H. Suzuki, “Multilinear polynomials and Frankl–Ray–Chaudhuri–Wilson type intersection theorems,” *Journal of Combinatorial Theory, Series A* **58**(2), 165–180 (1991). doi:10.1016/0097-3165(91)90058-O.
- [3] J. An and S. Lee, “Beyond the Positive Partial Transpose Squared Conjecture: The Qutrit Case,” preprint, arXiv:2607.15947v1 [quant-ph] (17 July 2026).

- [4] A. Arkhipov, “Extending and characterizing quantum magic games,” [arXiv:1209.3819](#) (2012).
- [5] P. Badziąg, K. Horodecki, M. Horodecki, J. Jenkinson, and S. J. Szarek, “Bound entangled states with extremal properties,” *Physical Review A* **90**, 012301 (2014).
[doi:10.1103/PhysRevA.90.012301](#). [arXiv:1309.7992v2](#).
- [6] Z. Bar-Yossef, T. S. Jayram, R. Kumar, and D. Sivakumar, “An information statistics approach to data stream and communication complexity,” *Journal of Computer and System Sciences* **68**(4), 702–732 (2004). [doi:10.1016/j.jcss.2003.11.006](#).
- [7] C. H. Bennett, G. Brassard, C. Crépeau, and U. M. Maurer, “Generalized privacy amplification,” *IEEE Transactions on Information Theory* **41**(6), 1915–1923 (1995).
[doi:10.1109/18.476316](#).
- [8] C. H. Bennett, G. Brassard, S. Popescu, B. Schumacher, J. A. Smolin, and W. K. Wootters, “Purification of noisy entanglement and faithful teleportation via noisy channels,” *Physical Review Letters* **76**, 722–725 (1996). [doi:10.1103/PhysRevLett.76.722](#). [arXiv:quant-ph/9511027](#).
- [9] C. H. Bennett, G. Brassard, and J.-M. Robert, “Privacy amplification by public discussion,” *SIAM Journal on Computing* **17**(2), 210–229 (1988). [doi:10.1137/0217014](#).
- [10] C. H. Bennett, D. P. DiVincenzo, T. Mor, P. W. Shor, J. A. Smolin, and B. M. Terhal, “Unextendible product bases and bound entanglement,” *Physical Review Letters* **82**, 5385–5388 (1999). [doi:10.1103/PhysRevLett.82.5385](#). The cited lemma locator refers to [arXiv:quant-ph/9808030v2](#).
- [11] C. H. Bennett, D. P. DiVincenzo, J. A. Smolin, and W. K. Wootters, “Mixed-state entanglement and quantum error correction,” *Physical Review A* **54**, 3824–3851 (1996).
[doi:10.1103/PhysRevA.54.3824](#). [arXiv:quant-ph/9604024](#).
- [12] L. Chen, Y. Yang, and W.-S. Tang, “Positive-partial-transpose square conjecture for $n = 3$,” *Physical Review A* **99**, 012337 (2019). [doi:10.1103/PhysRevA.99.012337](#).
- [13] G. Chiribella, K. R. Davidson, V. I. Paulsen, and M. Rahaman, “Positive maps and entanglement in real Hilbert spaces,” *Annales Henri Poincaré* **24**, 4139–4168 (2023).
[doi:10.1007/s00023-023-01325-x](#). [arXiv:2207.02510v2](#).
- [14] E. Chitambar, D. Leung, L. Mančinska, M. Ozols, and A. Winter, “Everything you always wanted to know about LOCC (but were afraid to ask),” *Communications in Mathematical Physics* **328**(1), 303–326 (2014). [doi:10.1007/s00220-014-1953-9](#). [arXiv:1210.4583](#).
- [15] M.-D. Choi, “Completely positive linear maps on complex matrices,” *Linear Algebra and its Applications* **10**, 285–290 (1975). [doi:10.1016/0024-3795\(75\)90075-0](#).
- [16] M. Christandl, A. Müller-Hermes, and M. M. Wolf, “When do composed maps become entanglement breaking?” *Annales Henri Poincaré* **20**, 2295–2322 (2019).
[doi:10.1007/s00023-019-00774-7](#). The cited section and theorem locators refer to [arXiv:1807.01266v2](#).
- [17] M. Curty, M. Lewenstein, and N. Lütkenhaus, “Entanglement as a precondition for secure quantum key distribution,” *Physical Review Letters* **92**, 217903 (2004).
[doi:10.1103/PhysRevLett.92.217903](#). [arXiv:quant-ph/0307151](#).
- [18] E. B. Davies and J. T. Lewis, “An operational approach to quantum probability,” *Communications in Mathematical Physics* **17**, 239–260 (1970). [doi:10.1007/BF01647093](#).

- [19] D. Deutsch, A. Ekert, R. Jozsa, C. Macchiavello, S. Popescu, and A. Sanpera, “Quantum privacy amplification and the security of quantum cryptography over noisy channels,” *Physical Review Letters* **77**, 2818–2821 (1996); erratum **80**, 2022 (1998). doi:10.1103/PhysRevLett.77.2818. Erratum DOI. arXiv:quant-ph/9604039.
- [20] I. Devetak and A. Winter, “Distillation of secret key and entanglement from quantum states,” *Proceedings of the Royal Society A* **461**, 207–235 (2005). doi:10.1098/rspa.2004.1372. arXiv:quant-ph/0306078.
- [21] S. N. Filippov, “Capacity of trace decreasing quantum operations and superadditivity of coherent information for a generalized erasure channel,” *Journal of Physics A: Mathematical and Theoretical* **54**, 255301 (2021). doi:10.1088/1751-8121/abfd61. The cited equation number refers to arXiv:2101.05686v3.
- [22] P. Frankl and V. Rödl, “Forbidden intersections,” *Transactions of the American Mathematical Society* **300**(1), 259–286 (1987). doi:10.1090/S0002-9947-1987-0871675-6.
- [23] P. Frankl and R. M. Wilson, “Intersection theorems with geometric consequences,” *Combinatorica* **1**(4), 357–368 (1981). doi:10.1007/BF02579457.
- [24] C. A. Fuchs and J. van de Graaf, “Cryptographic distinguishability measures for quantum-mechanical states,” *IEEE Transactions on Information Theory* **45**(4), 1216–1227 (1999). doi:10.1109/18.761271. arXiv:quant-ph/9712042.
- [25] N. Gisin, R. Renner, and S. Wolf, “Linking classical and quantum key agreement: Is there a classical analog to bound entanglement?” *Algorithmica* **34**(4), 389–412 (2002). doi:10.1007/s00453-002-0972-7.
- [26] N. Gisin and S. Wolf, “Linking classical and quantum key agreement: Is there ‘bound information’?” in *Advances in Cryptology—CRYPTO 2000*, Lecture Notes in Computer Science **1880**, pp. 482–500 (2000). doi:10.1007/3-540-44598-6_30. arXiv:quant-ph/0005042.
- [27] L. O. Hansen, A. Hauge, J. Myrheim, and P. Ø. Sollid, “Low-rank positive-partial-transpose states and their relation to product vectors,” *Physical Review A* **85**, 022309 (2012). doi:10.1103/PhysRevA.85.022309. See also arXiv:1104.1519v1.
- [28] E. P. Hanson, C. Rouzé, and D. Stilck França, “Eventually entanglement breaking Markovian dynamics: structure and characteristic times,” *Annales Henri Poincaré* **21**, 1517–1571 (2020). doi:10.1007/s00023-020-00906-4.
- [29] K. Horodecki, P. Źwikliński, A. Rutkowski, and M. Studziński, “On distilling secure key from reducible private states and (non)existence of entangled key-undistillable states,” *New Journal of Physics* **20**, 083021 (2018). arXiv:1612.08938v4.
- [30] K. Horodecki, M. Horodecki, P. Horodecki, and J. Oppenheim, “Secure key from bound entanglement,” *Physical Review Letters* **94**, 160502 (2005). doi:10.1103/PhysRevLett.94.160502. arXiv:quant-ph/0309110v2.
- [31] K. Horodecki, M. Horodecki, P. Horodecki, and J. Oppenheim, “General paradigm for distilling classical key from quantum states,” *IEEE Transactions on Information Theory* **55**(4), 1898–1929 (2009). doi:10.1109/TIT.2008.2009798. arXiv:quant-ph/0506189v2.
- [32] M. Horodecki, P. Horodecki, and R. Horodecki, “Mixed-state entanglement and distillation: Is there a ‘bound’ entanglement in nature?” *Physical Review Letters* **80**, 5239–5242 (1998). doi:10.1103/PhysRevLett.80.5239. arXiv:quant-ph/9801069.

- [33] M. Horodecki, P. W. Shor, and M. B. Ruskai, “Entanglement breaking channels,” *Reviews in Mathematical Physics* **15**, 629–641 (2003). doi:[10.1142/S0129055X03001709](https://doi.org/10.1142/S0129055X03001709). See also [arXiv:quant-ph/0302031v2](https://arxiv.org/abs/quant-ph/0302031v2).
- [34] P. Horodecki, “Separability criterion and inseparable mixed states with positive partial transposition,” *Physics Letters A* **232**, 333–339 (1997). doi:[10.1016/S0375-9601\(97\)00416-7](https://doi.org/10.1016/S0375-9601(97)00416-7). See also [arXiv:quant-ph/9703004v2](https://arxiv.org/abs/quant-ph/9703004v2).
- [35] A. Jamiołkowski, “Linear transformations which preserve trace and positive semidefiniteness of operators,” *Reports on Mathematical Physics* **3**, 275–278 (1972). doi:[10.1016/0034-4877\(72\)90011-0](https://doi.org/10.1016/0034-4877(72)90011-0).
- [36] O. Kallenberg, *Foundations of Modern Probability*, 2nd ed., Probability and Its Applications, Springer, New York (2002). doi:[10.1007/978-1-4757-4015-8](https://doi.org/10.1007/978-1-4757-4015-8).
- [37] M. Kennedy, N. A. Manor, and V. I. Paulsen, “Composition of PPT maps,” *Quantum Information and Computation* **18**(5–6), 472–480 (2018). doi:[10.26421/QIC18.5-6-4](https://doi.org/10.26421/QIC18.5-6-4). [arXiv:1710.08475v2](https://arxiv.org/abs/1710.08475v2). The cited theorem locator refers to the published version.
- [38] J. S. Kim and B. C. Sanders, “On the geometric distance between quantum states with positive partial transposition and private states,” *Letters in Mathematical Physics* **92**(1), 67–79 (2010). doi:[10.1007/s11005-010-0376-6](https://doi.org/10.1007/s11005-010-0376-6). [arXiv:0904.0295v1](https://arxiv.org/abs/0904.0295v1).
- [39] K. Kraus, “General state changes in quantum theory,” *Annals of Physics* **64**, 311–335 (1971). doi:[10.1016/0003-4916\(71\)90108-4](https://doi.org/10.1016/0003-4916(71)90108-4).
- [40] O. Krüger and R. F. Werner (editors), *Some open problems in quantum information theory* (2005). Problem 24, “Secret key from all entangled states,” p. 64. The cited problem locator refers to [arXiv:quant-ph/0504166v1](https://arxiv.org/abs/quant-ph/0504166v1).
- [41] U. M. Maurer, “Secret key agreement by public discussion from common information,” *IEEE Transactions on Information Theory* **39**(3), 733–742 (1993). doi:[10.1109/18.256484](https://doi.org/10.1109/18.256484).
- [42] N. D. Mermin, “Simple unified form for the major no-hidden-variables theorems,” *Physical Review Letters* **65**, 3373–3376 (1990). doi:[10.1103/PhysRevLett.65.3373](https://doi.org/10.1103/PhysRevLett.65.3373).
- [43] J. S. Milne, *Fields and Galois Theory*, version 5.10, September 2022. [Author’s text](#), Theorem 4.28 and Lemma 4.31, pp. 55–56.
- [44] S.-J. Park, “Every PPT channel has finite entanglement breaking index,” preprint, [arXiv:2608.13551v2](https://arxiv.org/abs/2608.13551v2) [quant-ph] (17 August 2026).
- [45] K. R. Parthasarathy, “On the maximal dimension of a completely entangled subspace for finite level quantum systems,” *Proceedings of the Indian Academy of Sciences (Mathematical Sciences)* **114**, 365–374 (2004). doi:[10.1007/BF02829441](https://doi.org/10.1007/BF02829441). See also [arXiv:quant-ph/0405077v1](https://arxiv.org/abs/quant-ph/0405077v1).
- [46] J. Pawels, N. Gisin, and R. Renner, “Bipartite Bound Information Exists,” preprint, [arXiv:2607.25838v2](https://arxiv.org/abs/2607.25838v2) [quant-ph] (8 September 2026).
- [47] A. Peres, “Incompatible results of quantum measurements,” *Physics Letters A* **151**, 107–108 (1990). doi:[10.1016/0375-9601\(90\)90172-K](https://doi.org/10.1016/0375-9601(90)90172-K).
- [48] A. Peres, “Two simple proofs of the Kochen–Specker theorem,” *Journal of Physics A: Mathematical and General* **24**(4), L175–L178 (1991). doi:[10.1088/0305-4470/24/4/003](https://doi.org/10.1088/0305-4470/24/4/003).

- [49] A. O. Pittenger, “Unextendible product bases and the construction of inseparable states,” *Linear Algebra and its Applications* **359**, 235–248 (2003). doi:[10.1016/S0024-3795\(02\)00423-8](https://doi.org/10.1016/S0024-3795(02)00423-8). See also [arXiv:quant-ph/0208028v1](https://arxiv.org/abs/quant-ph/0208028v1).
- [50] R. T. Powers and E. Størmer, “Free states of the canonical anticommutation relations,” *Communications in Mathematical Physics* **16**, 1–33 (1970). doi:[10.1007/BF01645492](https://doi.org/10.1007/BF01645492).
- [51] A. Prakash, J. Sikora, A. Varvitsiotis, and Z. Wei, “Completely positive semidefinite rank,” *Mathematical Programming* **171**(1–2), 397–431 (2018). doi:[10.1007/s10107-017-1198-4](https://doi.org/10.1007/s10107-017-1198-4). [arXiv:1604.07199](https://arxiv.org/abs/1604.07199).
- [52] M. Rahaman, S. Jaques, and V. I. Paulsen, “Eventually entanglement breaking maps,” *Journal of Mathematical Physics* **59**(6), 062201 (2018). doi:[10.1063/1.5024385](https://doi.org/10.1063/1.5024385). The cited theorem locator refers to [arXiv:1801.05542v2](https://arxiv.org/abs/1801.05542v2).
- [53] R. Renner and R. König, “Universally composable privacy amplification against quantum adversaries,” in *Theory of Cryptography—TCC 2005*, Lecture Notes in Computer Science **3378**, pp. 407–425 (2005). doi:[10.1007/978-3-540-30576-7_22](https://doi.org/10.1007/978-3-540-30576-7_22). [arXiv:quant-ph/0403133](https://arxiv.org/abs/quant-ph/0403133).
- [54] M. B. Ruskai, M. Junge, D. Kribs, P. Hayden, and A. Winter (organizers), *Operator structures in quantum information theory*, Banff International Research Station workshop report 12w5084 (2012). [Official report](#), Section 3, Problem G, p. 5.
- [55] J. Schliemann, D. Loss, and A. H. MacDonald, “Double-occupancy errors, adiabaticity, and entanglement of spin qubits in quantum dots,” *Physical Review B* **63**, 085311 (2001). doi:[10.1103/PhysRevB.63.085311](https://doi.org/10.1103/PhysRevB.63.085311). The cited equations refer to [arXiv:cond-mat/0009083v1](https://arxiv.org/abs/cond-mat/0009083v1).
- [56] S. Singh and I. Nechita, “The PPT² conjecture holds for all Choi-type maps,” *Annales Henri Poincaré* **23**, 3311–3329 (2022). doi:[10.1007/s00023-022-01166-0](https://doi.org/10.1007/s00023-022-01166-0). [arXiv:2011.03809v2](https://arxiv.org/abs/2011.03809v2).
- [57] F. Sottile, “From enumerative geometry to solving systems of polynomial equations,” in *Computations in Algebraic Geometry with Macaulay 2*, Algorithms and Computation in Mathematics **8**, Springer, pp. 101–129 (2001). [Author’s version](#), Sections 2.2–2.3, pp. 4–6.
- [58] The Stacks Project Authors, *The Stacks project*. [Tag 00FV](#) (Hilbert Nullstellensatz) and [Tag 0BBZ](#) (height theorem).